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Waddell

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(54) **FLEXIBLE ION GENERATOR DEVICE**

(71) Applicant: **Global Plasma Solutions, Inc.**,
Charlotte, NC (US)

(72) Inventor: **Charles Houston Waddell**, Roanoke,
VA (US)

(73) Assignee: **Global Plasma Solutions, Inc.**,
Charlotte, NC (US)

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This patent is subject to a terminal dis-
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Nov. 1, 2022, now Pat. No. 11,980,704, which is a
continuation of application No. 17/201,794, filed on
Mar. 15, 2021, now Pat. No. 11,511,014, which is a
continuation of application No. 16/785,995, filed on
Feb. 10, 2020, now Pat. No. 10,980,911, which is a
continuation-in-part of application No. 16/434,678,
filed on Jun. 7, 2019, now Pat. No. 10,695,455, which
is a continuation-in-part of application No.
15/816,027, filed on Nov. 17, 2017, now Pat. No.
10,322,205, which is a continuation of application
No. 15/412,825, filed on Jan. 23, 2017, now Pat. No.
9,849,208.

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21, 2016.

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H01J 33/02 (2006.01)
H05H 1/00 (2006.01)

(52) **U.S. Cl.**
CPC **A61L 9/22** (2013.01); **H01J 33/02**
(2013.01); **H05H 1/00** (2013.01); **A61L**
2209/16 (2013.01)

(58) **Field of Classification Search**
CPC **A61L 9/22**; **A61L 2209/16**; **H01J 33/02**;
H05H 1/00
See application file for complete search history.

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Primary Examiner — Wyatt A Stoffa

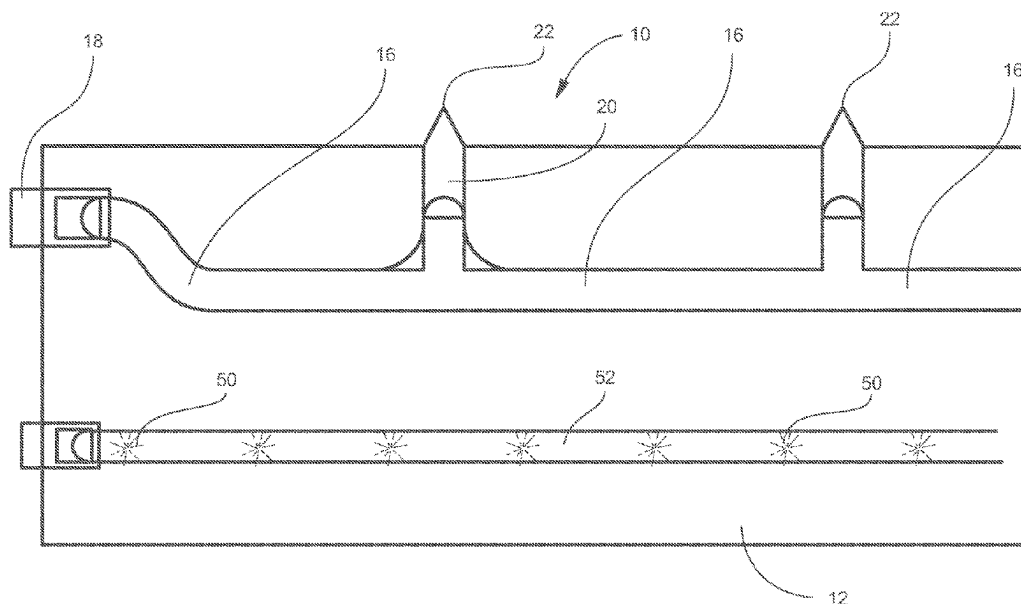
Assistant Examiner — Hanway Chang

(74) *Attorney, Agent, or Firm* — Seth L. Hudson;
Maynard Nexsen PC

(57) **ABSTRACT**

A flexible ion generator device that includes a dielectric
layer having a first end, a second end, a first side, a second
side, a top side, and a bottom side, at least one trace
positioned on the dielectric layer and having a plurality of
emitters engaged to the at least one trace. A plurality of lights
disposed on the dielectric layer.

20 Claims, 14 Drawing Sheets



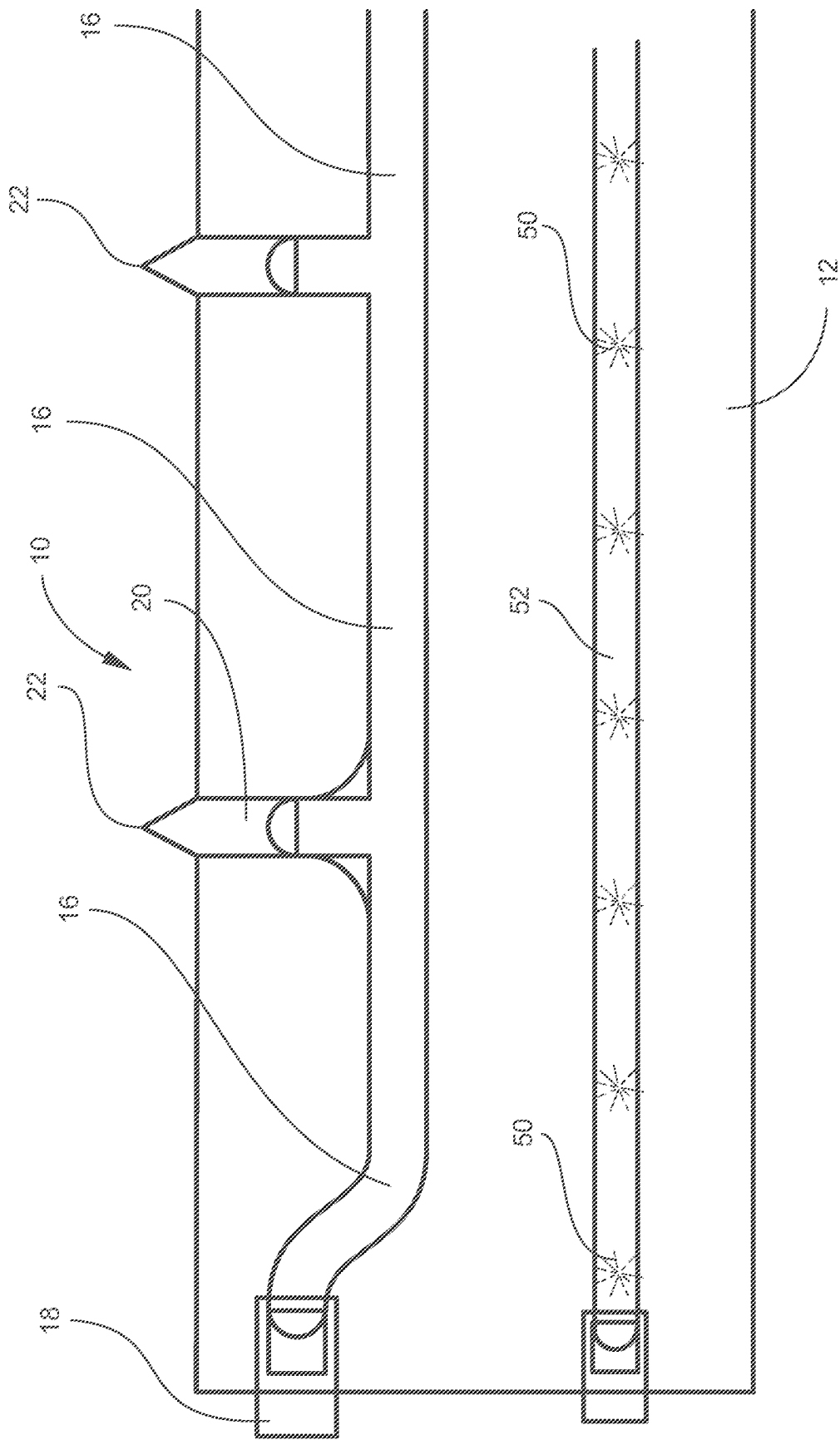


FIG. 1A

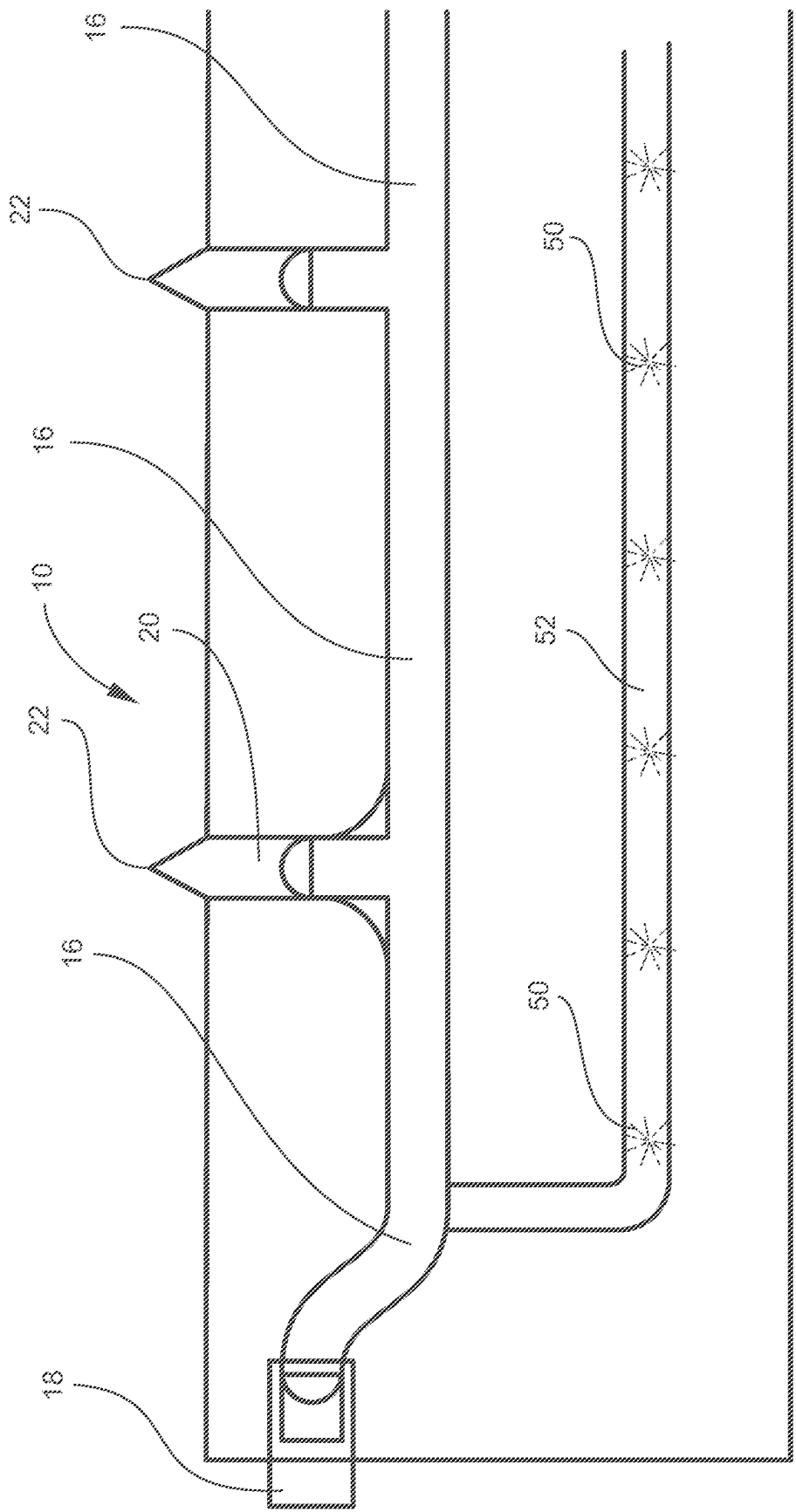


FIG. 1B

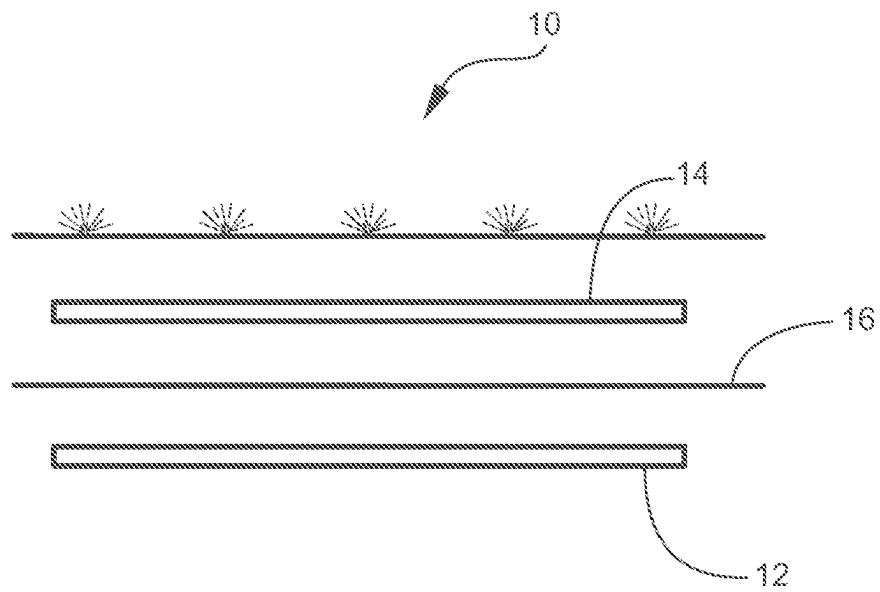


FIG. 2

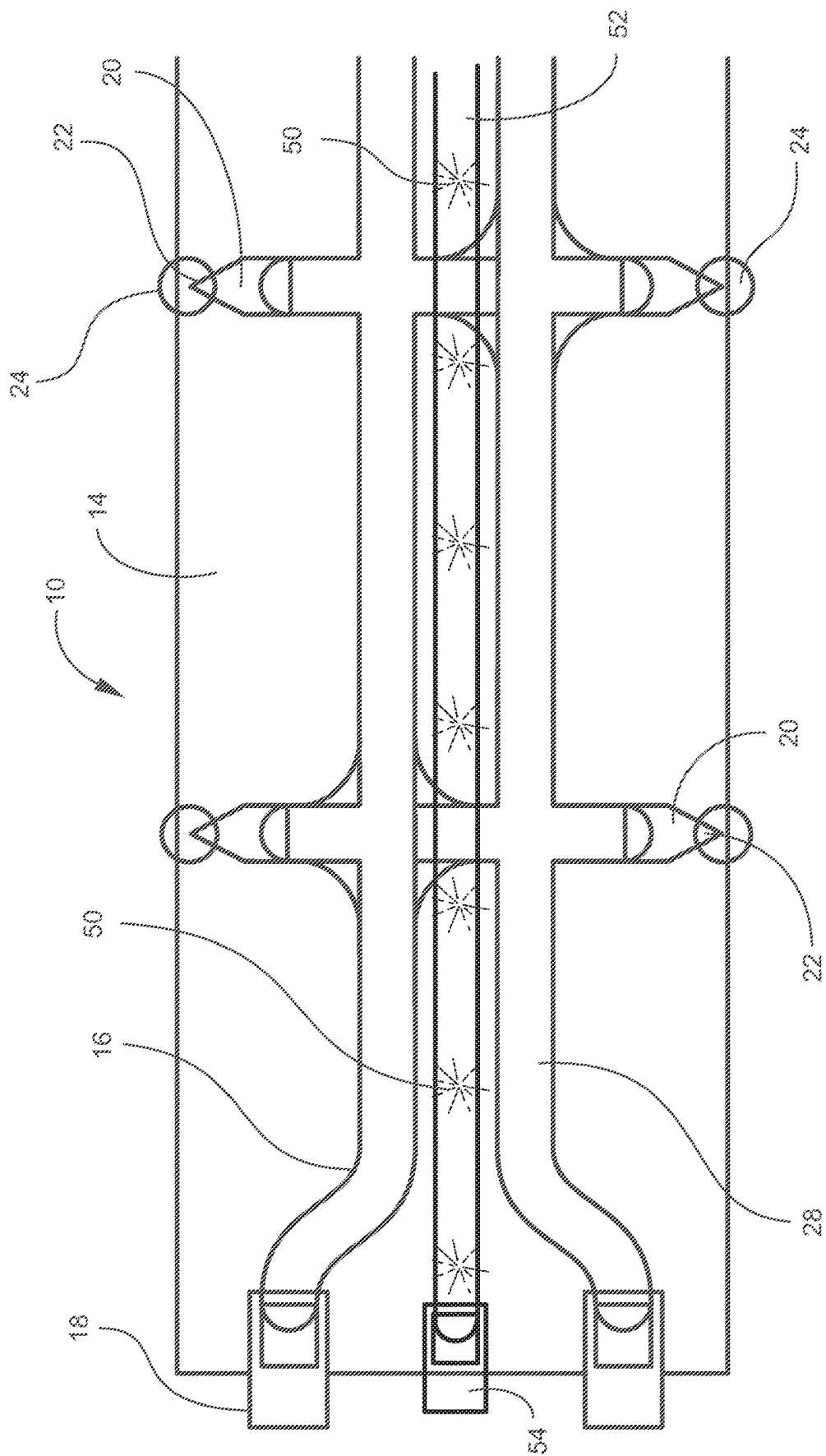


FIG. 3

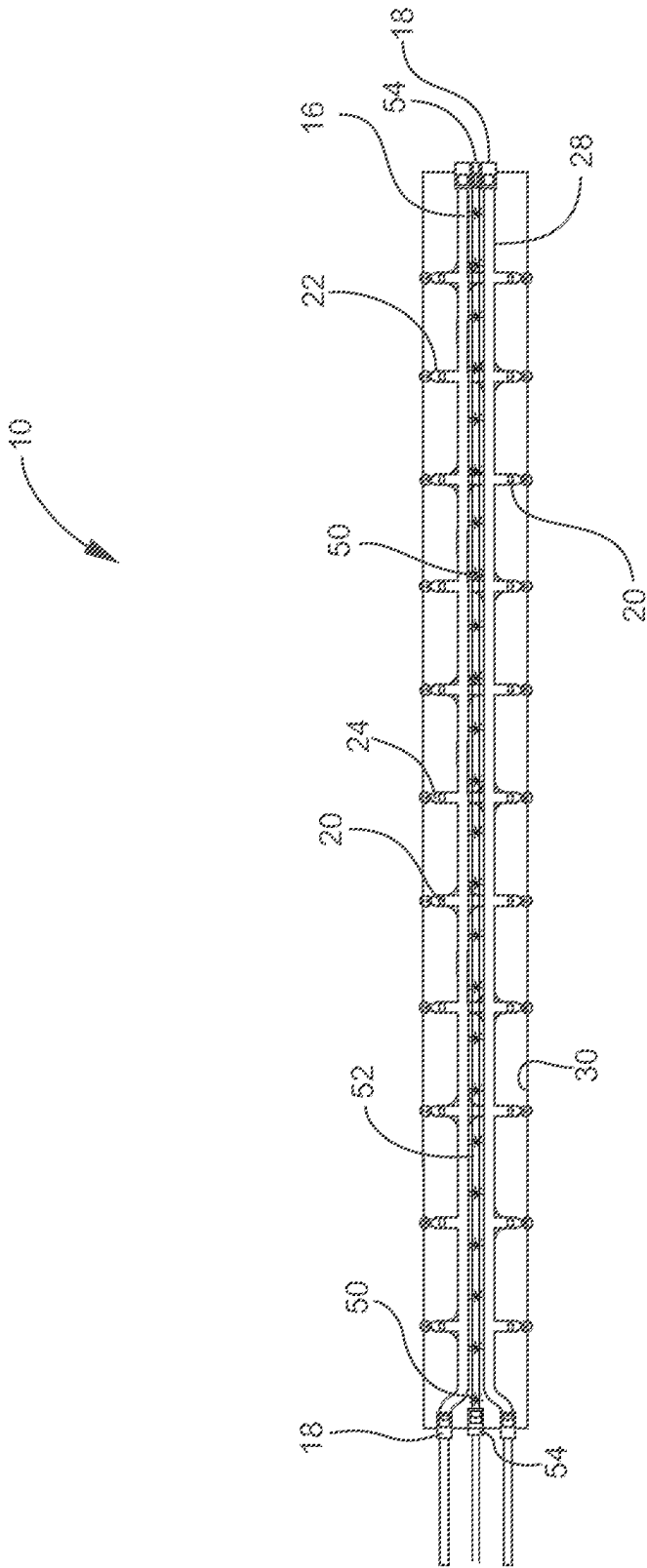
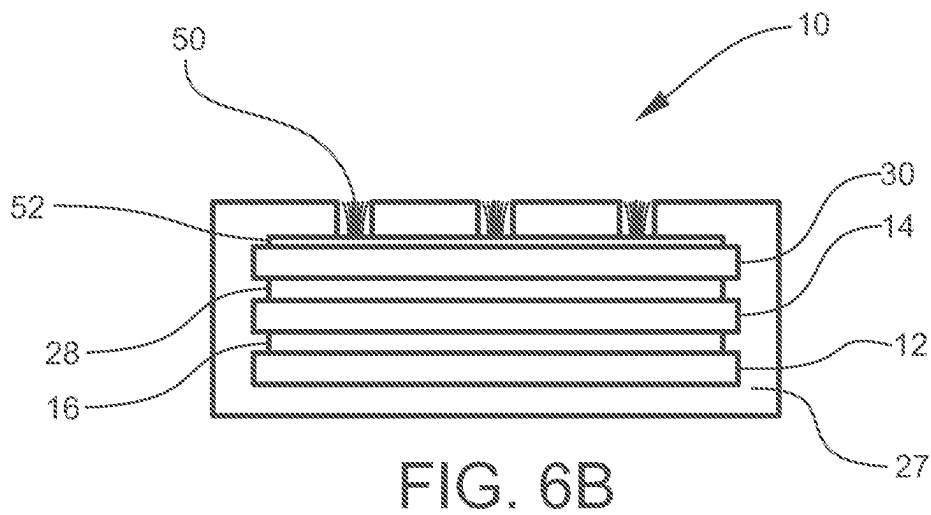
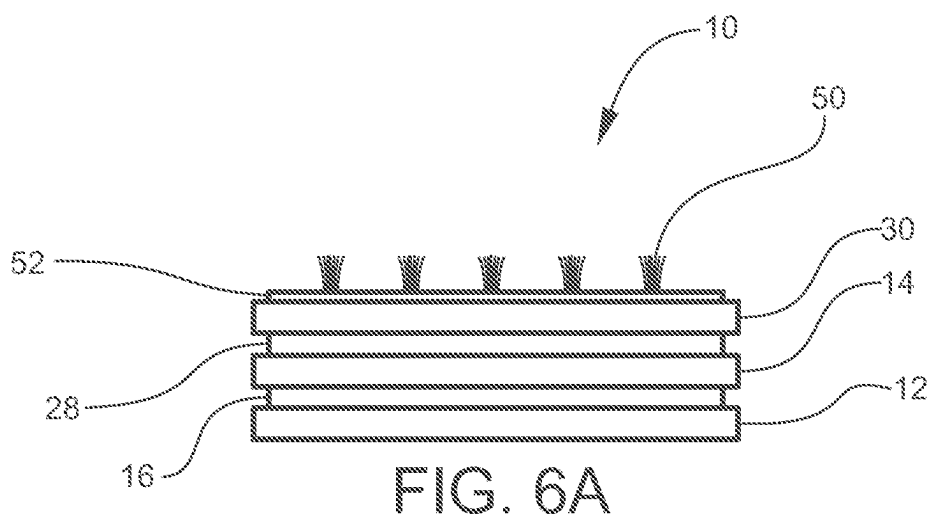
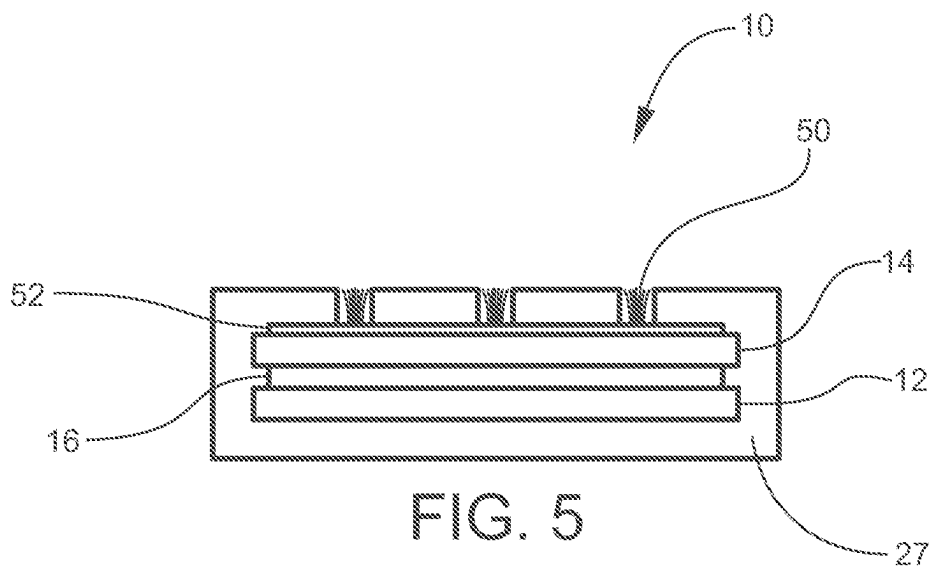


FIG. 4



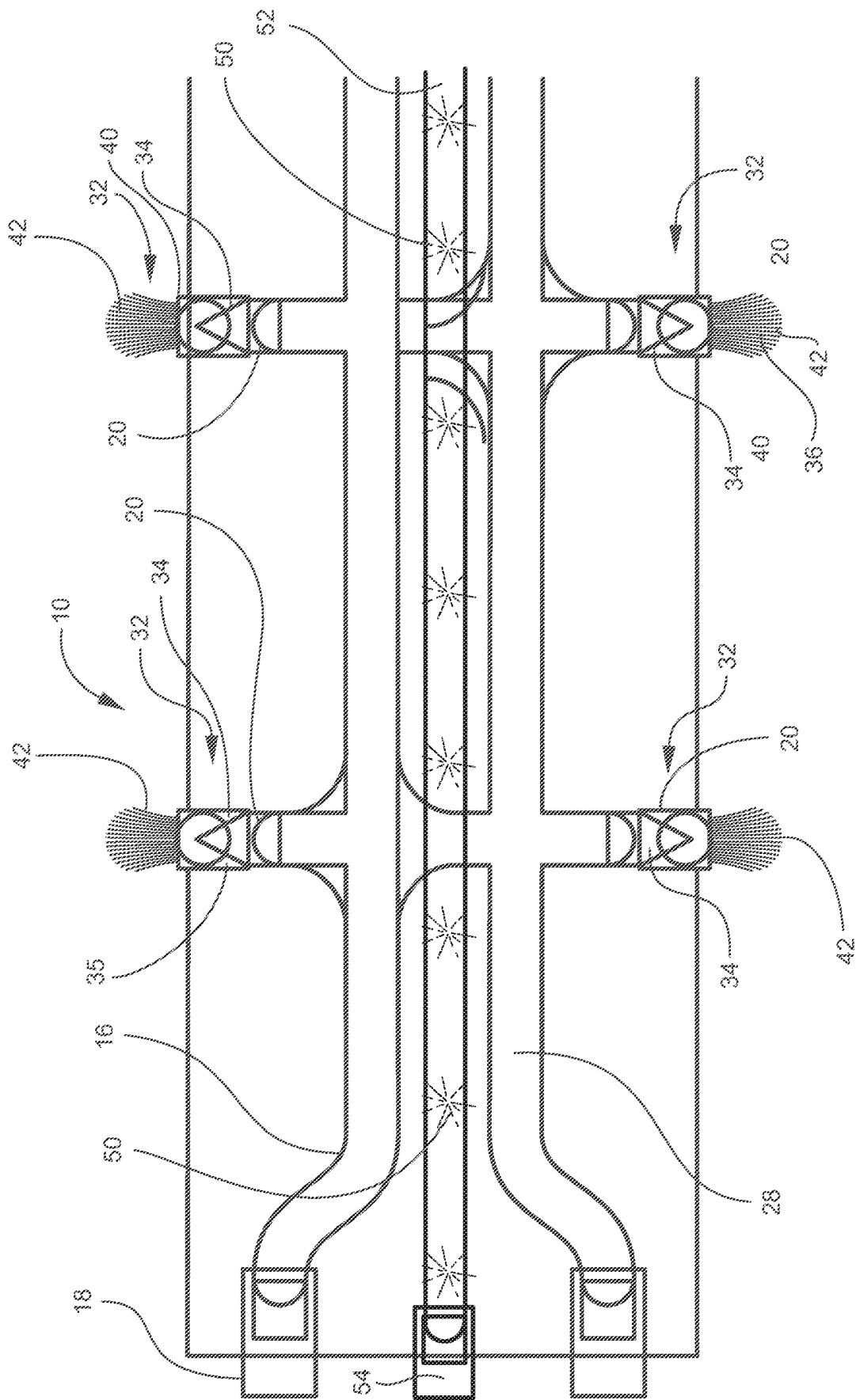


FIG. 7

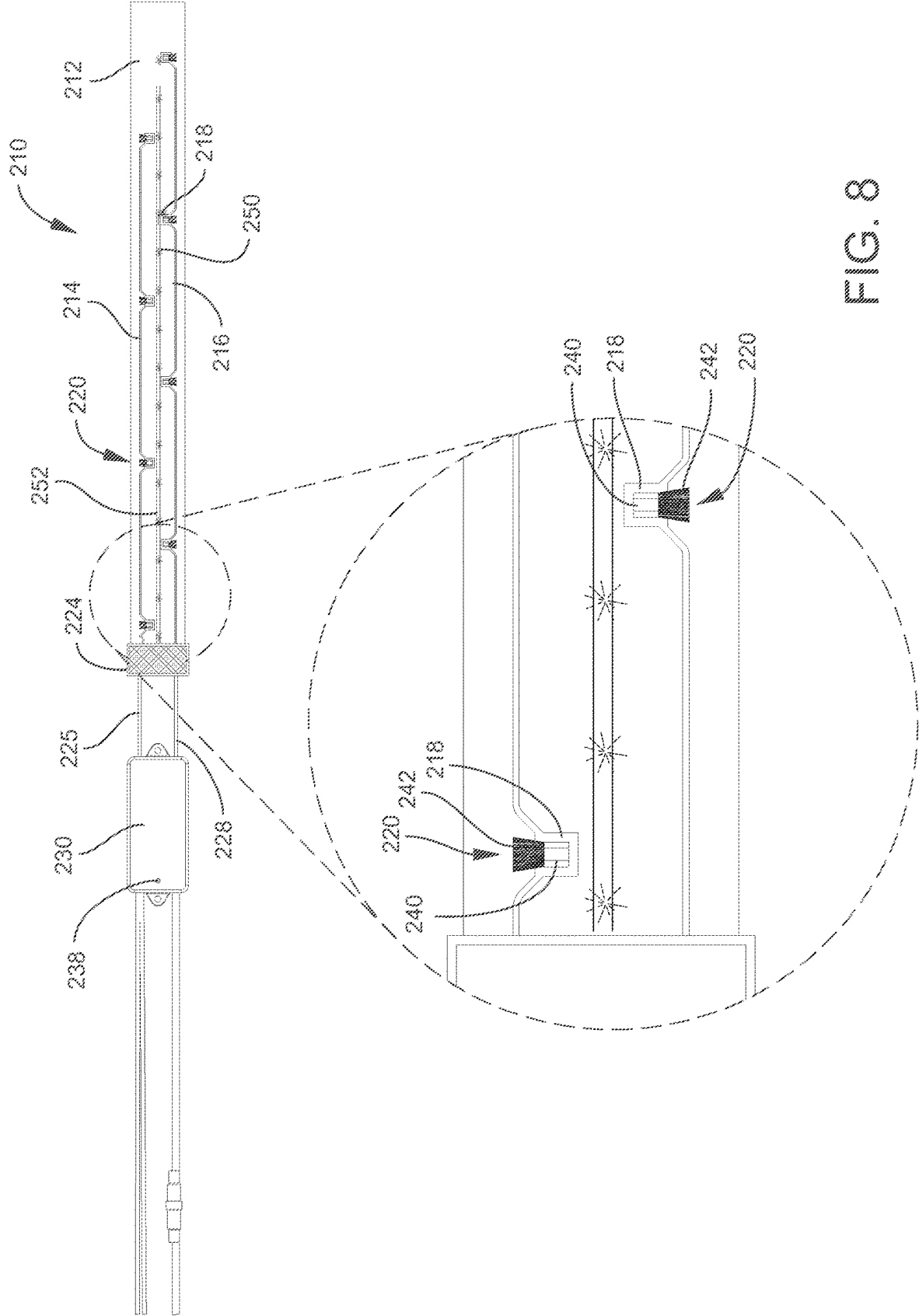


FIG. 8

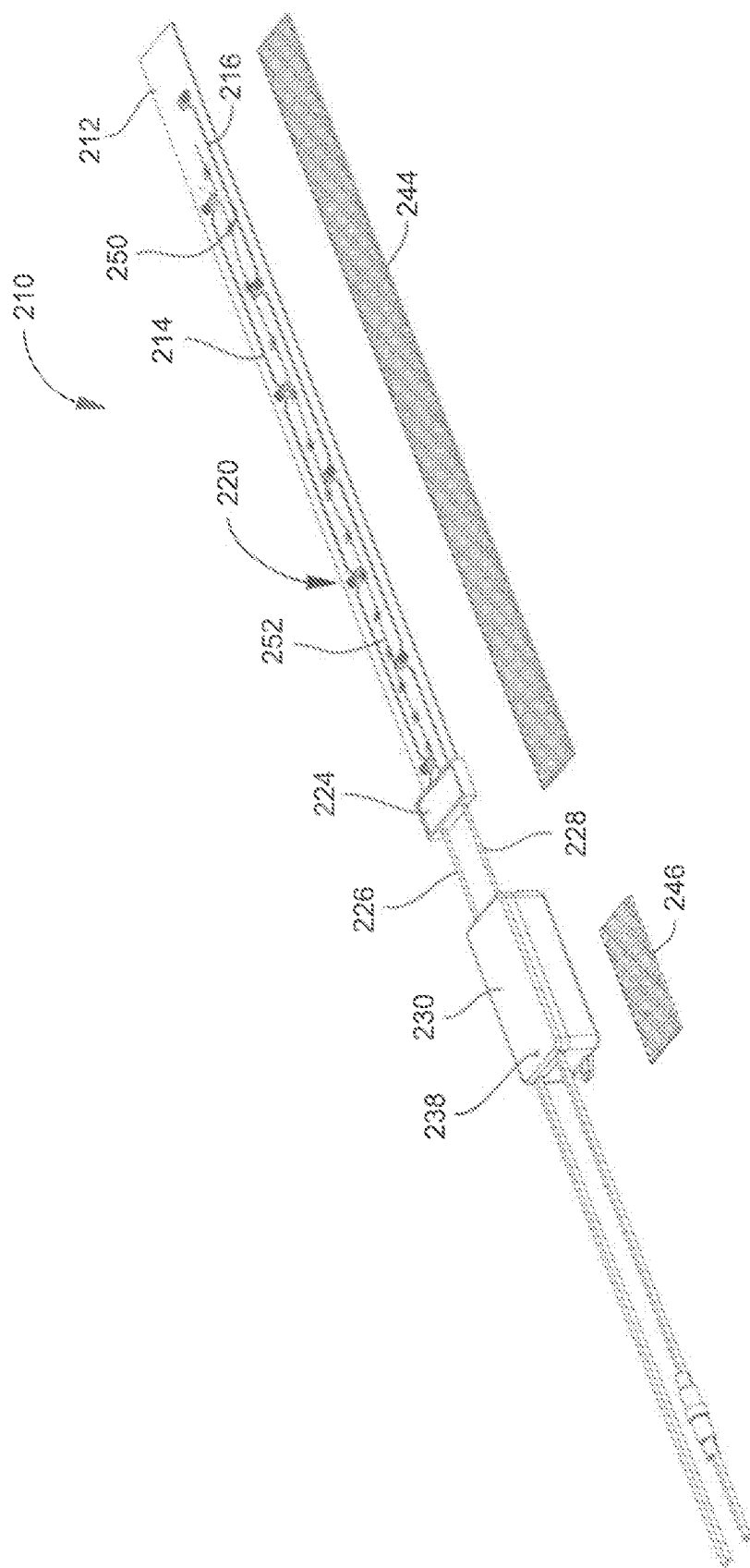


FIG. 9

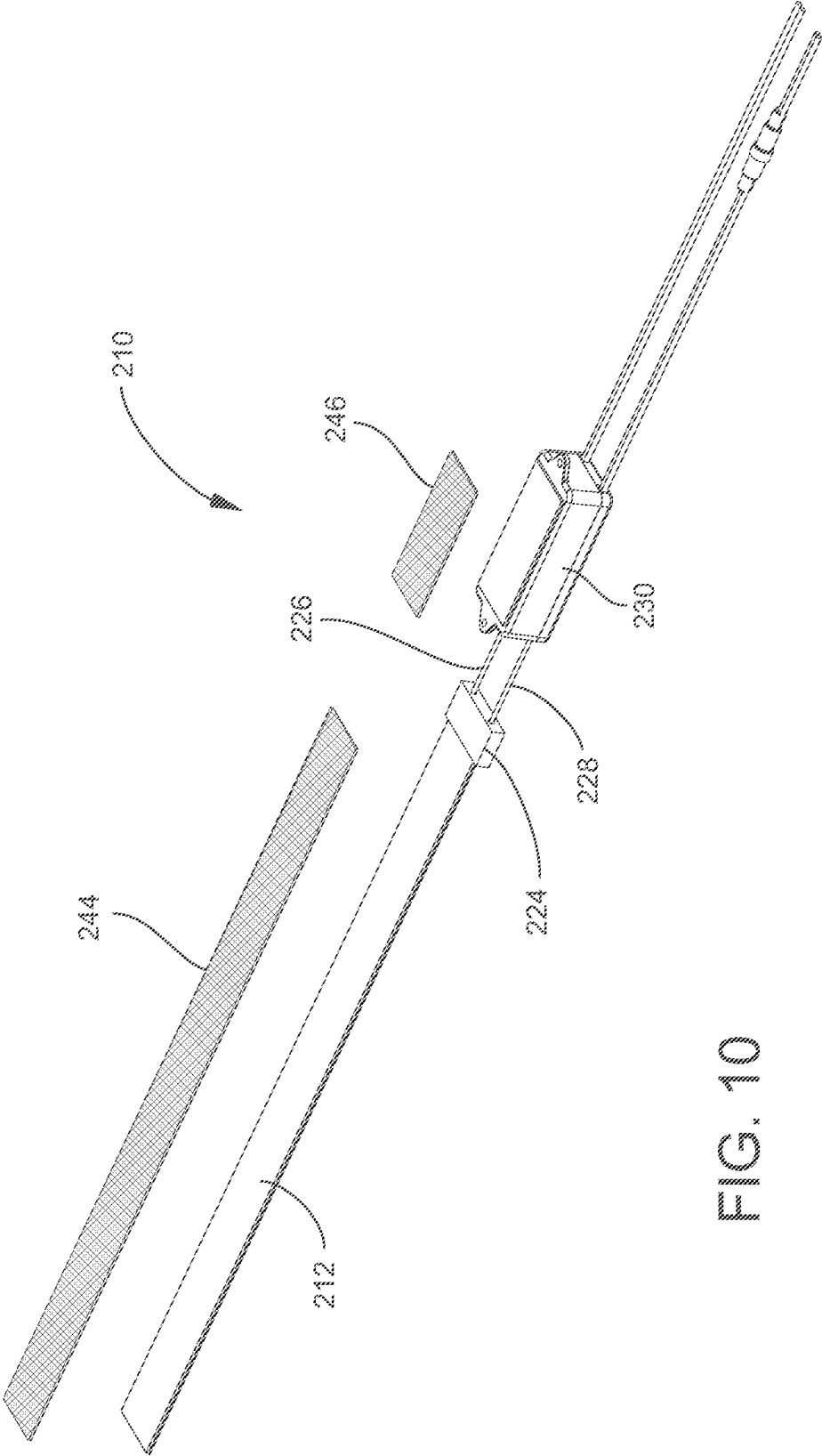
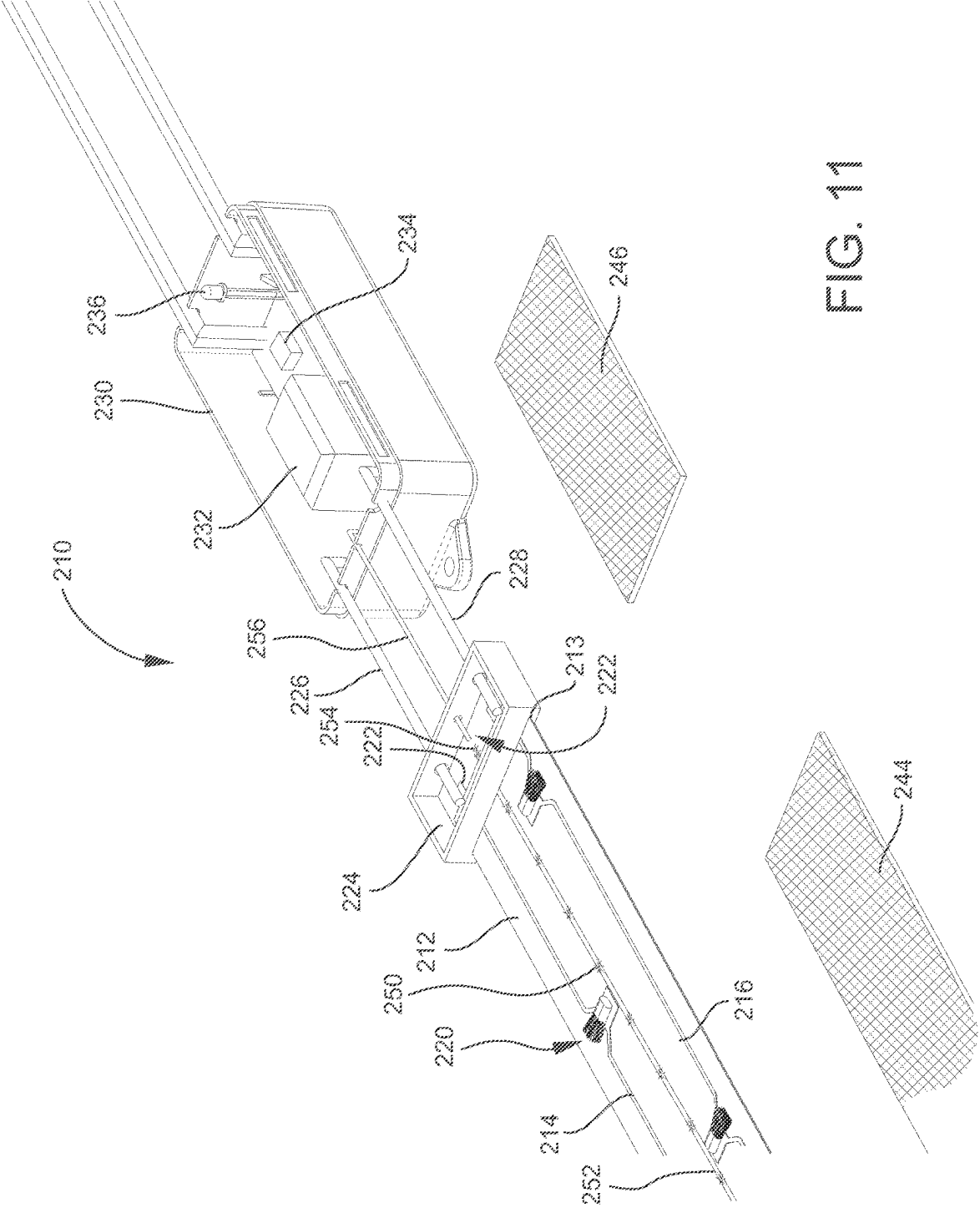


FIG. 10



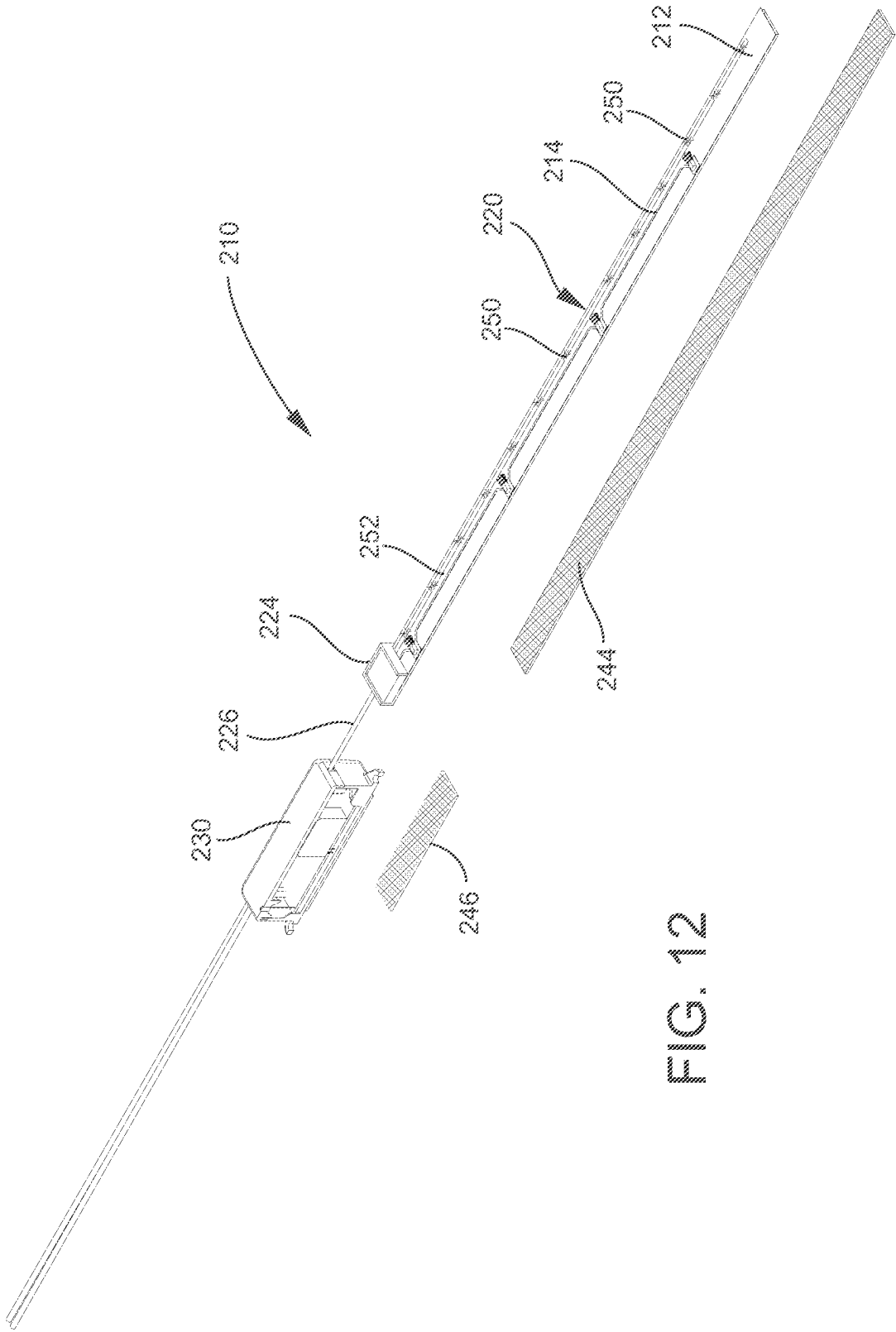
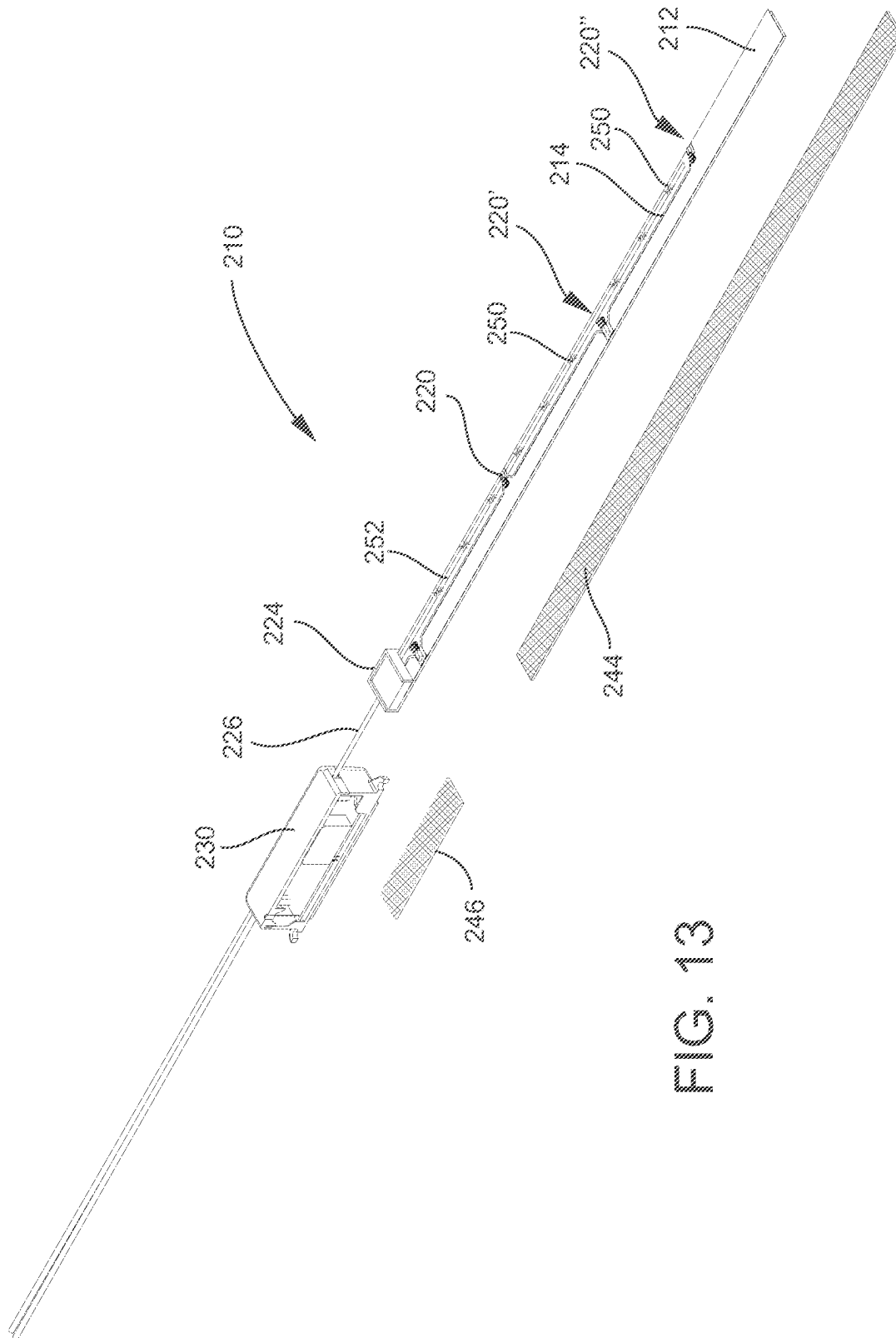


FIG. 12





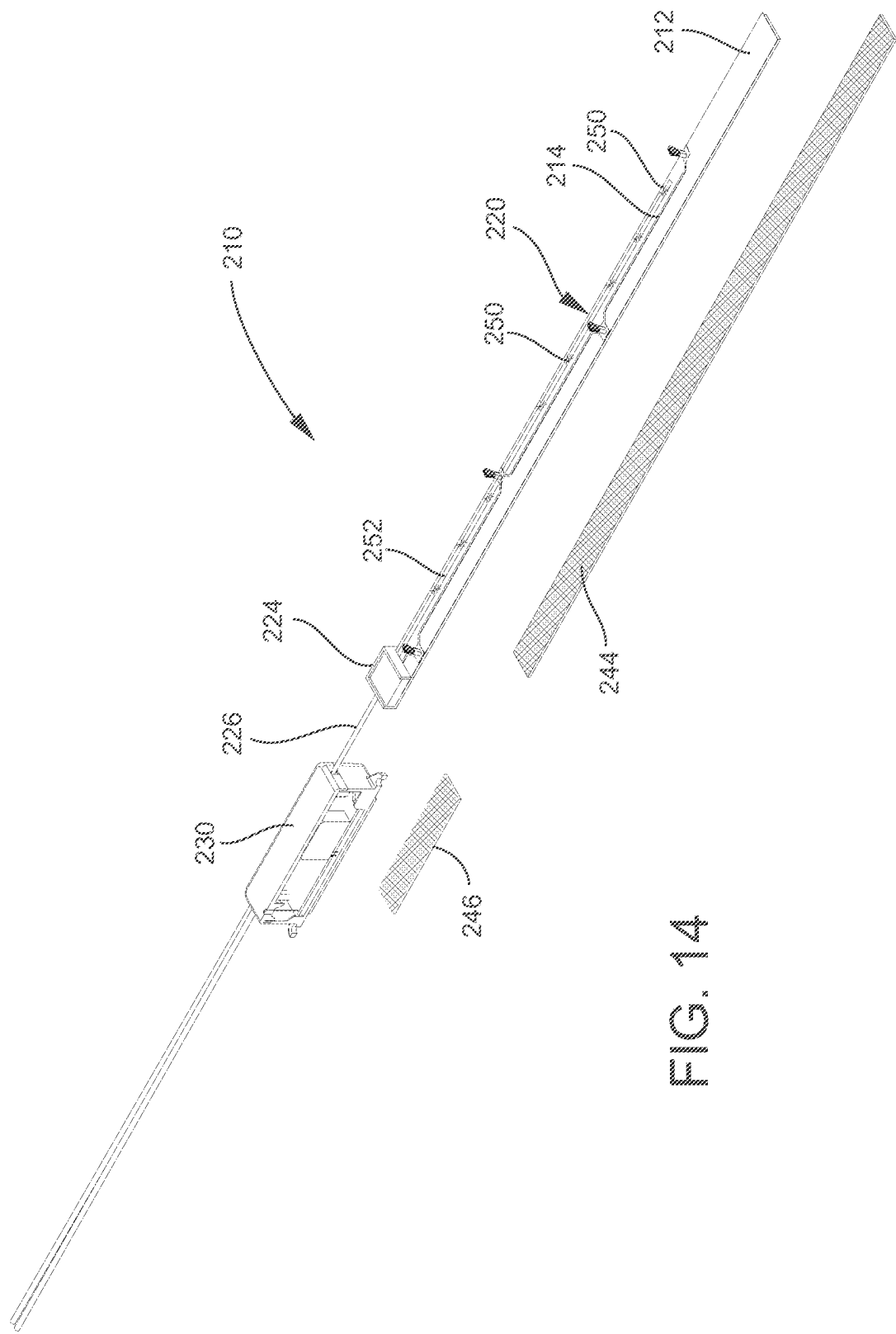


FIG. 14

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FLEXIBLE ION GENERATOR DEVICE**CROSS REFERENCE TO RELATED PATENT APPLICATION**

This application is a continuation application of U.S. patent Ser. No. 17/978,622, filed Nov. 1, 2022, and entitled “FLEXIBLE ION GENERATOR DEVICE”, which is a continuation application of U.S. patent Ser. No. 17/201,794, filed on Mar. 15, 2021, and entitled “FLEXIBLE ION GENERATOR DEVICE”, which is a continuation of U.S. Pat. No. 10,980,911, issued on Apr. 20, 2021, and entitled “FLEXIBLE ION GENERATOR DEVICE”, which is a continuation-in-part of U.S. Pat. No. 10,695,455, issued on Jun. 30, 2020, and entitled “FLEXIBLE ION GENERATOR DEVICE”, which is a continuation-in-part of U.S. Pat. No. 10,322,205, issued Jun. 18, 2019, and entitled “FLEXIBLE ION GENERATION DEVICE”, which is a continuation of U.S. Pat. No. 9,849,208 issued on Dec. 26, 2017 and entitled “FLEXIBLE ION GENERATION DEVICE”, which claims the benefit of U.S. Provisional Patent Application No. 62/281,318, filed on Jan. 21, 2016, and entitled “FLEXIBLE ION ELECTRODE,” the contents of which are incorporated in full by reference herein.

FIELD OF THE INVENTION

The present invention relates generally to the field of air treatment, and more particularly to the treatment of air using ionization that is produced using a flexible ion generation device for dispersing the ions into the surrounding air and containing UV lights to further sanitize the surrounding air and adjacent surfaces.

BACKGROUND OF THE INVENTION

Air and other fluids are commonly treated and delivered for a variety of applications. For example, in heating, ventilation and air-conditioning (HVAC) applications, air may be heated, cooled, humidified, dehumidified, filtered or otherwise treated for delivery into residential, commercial or other spaces.

Needs exist for improved systems and methods of treating and delivering purified air for these and other applications, including sanitizing surrounding air and adjacent surfaces. It is to the provision of improved systems and methods meeting these needs that the present invention is primarily directed.

BRIEF SUMMARY OF THE INVENTION

According to an embodiment of the present invention, the flexible ion generator device includes one dielectric layer, at least one trace having a first end and a second end. The at least one trace is engaged to the dielectric layer, and at least one emitter engaged to the trace for emitting ions. At least one UV light is disposed on the dielectric layer.

According to another embodiment of the present invention, the flexible ion generator device includes a conductive wire disposed on the dielectric layer and that at least one UV light is engaged to the conductive wire.

According to yet another embodiment of the present invention, the flexible ion generator device includes a conductive wire disposed on the dielectric layer and a plurality of UV lights engaged to the conductive wire.

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According to yet another embodiment of the present invention, the flexible ion generator device includes a conductive wire disposed on the dielectric layer and substantially parallel with the trace.

According to yet another embodiment of the present invention, the flexible ion generator device includes at least one trace positioned in close proximity to the first side of the dielectric layer and generally is parallel with the first side of the dielectric layer. At predetermined distances along the length of the at least one trace, the at least one trace extends downwardly towards the second side of the dielectric layer to a first side of the contact point and then from a second side of the contact point, the at least one trace extends towards the top side of the dielectric layer and extends generally parallel with the top side of the dielectric layer.

According to yet another embodiment of the present invention, the flexible ion generator device includes a coupler having a base that extends to an outer edge and a first pair of opposed sidewalls and a second pair of opposed sidewalls extend upwardly from the outer edge to an upper edge, forming a cavity therein. A top portion is disposed on the upper edge, and a slot is formed in one of the sidewalls extending from the external surface to the internal surface for receiving the first end of the dielectric layer.

According to yet another embodiment of the present invention, the flexible ion generator device includes an attachment device disposed on the bottom side of the dielectric layer.

According to yet another embodiment of the present invention, the flexible ion generator device includes a dielectric layer having a first end, a second end, a first side, a second side, a top side, and a bottom side. At least one trace positioned on the dielectric layer and has a plurality of emitters engaged to the at least one trace, wherein the trace extends along the top side of the dielectric layer along a substantially parallel plane with respect to either the first side or the second side and in predetermined locations periodically along the length of the dielectric layer, the at least one trace extends downwardly from the parallel plane for a distance and then upwardly towards the parallel plane.

According to yet another embodiment of the present invention, the flexible ion generator device wherein the dielectric layer may be cut anywhere along its length without affecting the operating of the device.

According to yet another embodiment of the present invention, the flexible ion generator device wherein at least one emitter extends upwardly from the dielectric layer.

According to yet another embodiment of the present invention, the flexible ion generator device wherein the emitters face toward either the first side or the second side in an alternating arrangement.

According to yet another embodiment of the present invention, the flexible ion generator device includes a power supply device engaged to the flexible ion generator device.

According to yet another embodiment of the present invention, the flexible ion generator device includes a second trace engaged to the top portion of the second dielectric layer and a third dielectric layer having a top portion and a bottom portion, wherein the bottom portion of the third dielectric layer is engaged to the second trace and the top portion of the second dielectric layer.

According to yet another embodiment of the present invention, the flexible ion generator device includes a dielectric layer having a first end, a second end, a first side, a second side, a top side, and a bottom side. A trace positioned on the dielectric layer and having a plurality of emitters engaged to the trace, wherein the trace extends

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along the top side of the dielectric layer and along a substantially parallel plane with respect to either the first side or the second side and in predetermined locations periodically along the length of the dielectric layer, the trace extends downwardly from the parallel plane for a distance and then upwardly towards the parallel plane.

According to yet another embodiment of the present invention, the flexible ion generator device includes a plurality of contact points along the trace for receiving the plurality of emitters.

According to yet another embodiment of the present invention, the flexible ion generator devices includes a coupler having a base that extends to an outer edge and a first pair of opposed sidewalls and a second pair of opposed sidewalls extend upwardly from the outer edge to an upper edge, forming a cavity therein, a top portion is disposed on the upper edge, a slot is formed in one of the sidewalls extending from the external surface to the internal surface for receiving the first end of the dielectric layer.

BRIEF DESCRIPTION OF THE DRAWINGS

The present invention is illustrated and described herein with reference to the various drawings, in which like reference numbers denote like method steps and/or system components, respectively, and in which:

FIG. 1*a* is a top view of one embodiment of the flexible ion generation device;

FIG. 1*b* is a top view of another embodiment of the flexible ion generator device;

FIG. 2 is an exploded view of one embodiment of the flexible ion generator device;

FIG. 3 is a top view of another embodiment of the flexible ion generator device;

FIG. 4 is a top view of another embodiment of the flexible ion generator device;

FIG. 5 is a cross-sectional view of another embodiment of the flexible ion generator device;

FIG. 6*a* is a cross-sectional view of another embodiment of the flexible ion generator device;

FIG. 6*b* is a cross-sectional view of another embodiment of the flexible ion generator device;

FIG. 7 is a top perspective view of another embodiment of the flexible ion generator device;

FIG. 8 is a top perspective view of an embodiment of the flexible ion generator device with an enlarged portion;

FIG. 9 is a top perspective view of an embodiment of the flexible ion generator device;

FIG. 10 is a bottom perspective view of an embodiment of the flexible ion generator device;

FIG. 11 is a top perspective view of an embodiment of the flexible ion generator device;

FIG. 12 is a top perspective view of an embodiment of the flexible ion generator device;

FIG. 13 is a top perspective view of an embodiment of the flexible ion generator device; and

FIG. 14 is a top perspective view of an embodiment of the flexible ion generator device.

DETAILED DESCRIPTION OF THE INVENTION

The present invention may be understood more readily by reference to the following detailed description of the invention taken in connection with the accompanying drawing figures, which form a part of this disclosure. It is to be understood that this invention is not limited to the specific

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devices, methods, conditions or parameters described and/or shown herein, and that the terminology used herein is for the purpose of describing particular embodiments by way of example only and is not intended to be limiting of the claimed invention. Any and all patents and other publications identified in this specification are incorporated by reference as though fully set forth herein.

Also, as used in the specification including the appended claims, the singular forms “a,” “an,” and “the” include the plural, and reference to a particular numerical value includes at least that particular value, unless the context clearly dictates otherwise. Ranges may be expressed herein as from “about” or “approximately” one particular value and/or to “about” or “approximately” another particular value. When such a range is expressed, another embodiment includes from the one particular value and/or to the other particular value. Similarly, when values are expressed as approximations, by use of the antecedent “about,” it will be understood that the particular value forms another embodiment.

Referring now specifically to the drawings, a flexible ion generator device is described herein and illustrated in FIGS. 1*a*, 1*b*, and 2 and is shown generally at reference numeral 10. The flexible ion generator device 10 has at least a first dielectric layer 12 and, optionally, a second dielectric layer 14. The first dielectric layer 12 contains a top portion, a bottom portion, a top side, a bottom side, a left side, and a right side, wherein the top side and the bottom side are opposed to each other and the left side and the right side are opposed to each other. The second dielectric layer 14 contains a top portion, a bottom portion, a top side, a bottom side, a left side, and a right side, wherein the top side and the bottom side are opposed to each other and the left side and the right side are opposed to each other.

The flexible ion generator device 10 includes a first dielectric layer 12 having a trace 16 disposed thereon. As illustrated in FIGS. 1*a* and 1*b*, the trace 16 is disposed on the top portion of the first dielectric layer 12. In another embodiment, as shown in FIG. 2, the flexible ion generator device 10 includes a first dielectric layer 12 and a second dielectric layer 14 having a trace 16 disposed therebetween (the trace 16 will be considered a first trace when more than one trace is utilized with two or more dielectric layers). In this embodiment, the trace 16 is adjacent to the top portion of the first dielectric layer 12 and the bottom side of the second dielectric layer 14, and may be engaged to the top portion of the first dielectric layer 12 or engaged to the bottom side of the second dielectric layer 14.

The trace 16 has a first end and a second end. The flexible ion generator device 10 may have one or more traces 16, such as copper traces, positioned on the top portion of the first dielectric layer 12 or between the first dielectric layer 12 and the second dielectric layer 14, wherein the trace 16 is adjacent to the top portion of the first dielectric layer 12 and the bottom side of the second dielectric layer 14 and may be engaged to the top portion of the first dielectric layer 12 or engaged to the bottom side of the second dielectric layer 14. Alternatively, the trace 16 may be composed of other conducting materials such as brass, stainless steel, titanium, gold, silver, tungsten, carbon, mixtures thereof, and the like. In the embodiment consisting of a first dielectric layer 12 and a second dielectric layer 14, the bottom portion of the second dielectric layer 14 may be formed over the trace 16 and coupled to the first dielectric layer 12. It will be appreciated that while the trace 16 as illustrated in FIGS. 1*a* and 1*b* is positioned on the top portion of the first dielectric layer 12, the trace 16 may also be positioned on the bottom portion of the second dielectric layer 14, wherein the first

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dielectric layer 12 is formed over the trace 16 and coupled to the second dielectric layer 14.

As illustrated in FIGS. 1a and 1b, the trace 16 extends longitudinally along the length of the flexible ion generator device 10. In other words, the trace 16 extends from the left side to the right side of the first dielectric layer 12 and the optional second dielectric layer 14. The trace 16 contains a first end and a second end, whereby the first end is disposed adjacent the right side of the first dielectric layer 12 and second dielectric layer 14 and the second end is disposed adjacent the left side of the first dielectric layer 12 and the second dielectric layer 14. The first end and the second end of the trace 16 may extend beyond the right side and left side of the first dielectric layer 12 and the second dielectric layer 14. A conductive pad or connector 18 may be disposed on the first end and/or the second end of the trace 16. As illustrated in FIGS. 1a and 1b, the connector 18 is disposed on the second end of the trace 16. The connector 18 is engaged to a power supply for supplying power to the flexible ion electrode 10, and more specifically the trace 16. The embodiment illustrated in FIG. 2 will look the same as the embodiment in FIGS. 1a and 1b, except the second dielectric layer 14 is engaged to the first dielectric layer 12.

As shown in FIGS. 1a, 1b, and 2, at least one light 50 may be disposed on the first dielectric layer 12, and preferably two or more lights 50 are disposed on the first dielectric layer 12, and most preferably, a plurality of lights 50 are disposed on the first dielectric layer 12. As illustrated, the lights 50 are disposed on the top portion of the first dielectric layer 12 but may also be disposed on the bottom portion of the first dielectric layer 12. A conductive wire 52 is disposed along the first dielectric layer 12 and each light 50 contacts the conductive wire 52. As illustrated in FIGS. 1a and 1b, the conductive wire 52 is disposed on the top portion of the first dielectric layer 12 and may be parallel to the trace 16. Alternatively, the conductive wire 52 may be positioned on the bottom portion of the first dielectric layer 12 or within the first dielectric layer 12. The conductive wire 52 extends along the length of the first dielectric layer 12 and each light 50 is engaged to the conductive wire 52. The conductive wire 52 provides power to the lights 50. A connector 54 may be engaged to an end of the conductive wire 52 for engagement to a power supply for providing power to the conductive wire 52 and ultimately to the lights 50. Alternatively, the conductive wire 52 may be engaged to a connector 54 (FIG. 1a) or the trace 16 (FIG. 1b), and either the connector 54 or trace 16 provide the power that is ultimately transferred to the lights 50, enabling the lights 50 to illuminate. The conductive wire 52 may extend from the first end of the first dielectric layer 12 to the second end of the first dielectric layer 12. The conductive wire 52 is composed of metal that can conduct electricity, such as copper or aluminum.

An emitter 20 may be engaged or etched into the trace 16. As illustrated in FIGS. 1 and 3, the emitter 20 extends to at least the first side and/or second side of the flexible ion electrode 10. The emitter 20 extends to the top side and/or the bottom side of the first dielectric layer 12 (FIGS. 1a and 1b) and the second dielectric layer 14 (FIG. 3). The emitter 20 extends from the trace 16 to either the top side or bottom side of the first dielectric layer 12 and the second dielectric layer 14. All emitters 20 of the trace can extend to the same side of the dielectric layer (12, 14), may alternate sides of the dielectric layer (12, 14), or may extend to different sides of the dielectric layer (12, 14) at different intervals. The first end of the emitter 20 may be etched into the trace 16 and the second end of the emitter 20 extends to a point 22. As

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illustrated, the second end of the emitter 20 has a gradually reducing width that terminates at a sharp point 22, allowing ions to flow therefrom.

The point 22 of the emitter 20 is not connected, coupled, or engaged to the first dielectric layer 12, or the optional second dielectric layer 14 and extends outwards from these dielectric layers (12, 14). In other words, the emitter 20 and/or point 22 may extend beyond the top side or bottom side of the first dielectric layer 12 and/or the optional second dielectric layer 14. The point 22 may be coated or plated with a corrosion resistant layer such as gold or other coating material.

The point 22 is disposed on the second end of the emitter 20 and allows ions to flow therefrom. Each trace 16 may contain at least one emitter 20, preferably at least two emitters 20, and more preferably a plurality of emitters 20. It is important to note that the first dielectric layer 12 and the optional second dielectric layer 14 does not cover the point 22, thus allowing ions to flow from the point 22 and into the surrounding area. The dielectric layers (12, 14) adjacent the point may be cut-away, pulled back, or otherwise removed, allowing ions to freely flow from the end of the point 22.

In another alternative embodiment, only one dielectric layer (12, 14) may be cut-way, pulled back, or otherwise removed from the point 22 of the emitter. For example and as illustrated in FIG. 3, the trace 16 and emitter 20 are disposed on the top portion of the first dielectric layer 12, and the second dielectric layer 14 is formed over the trace 16, as shown in FIG. 2, and the emitter 20 is disposed on the first dielectric layer 12. The point 22 of the emitter 20 does not extend outward from the first dielectric layer 12 and second dielectric layer 14. Instead, the portion of the second dielectric layer that would be cover or be over top the point 22 is cut-away, pulled back, or otherwise removed, forming an opening 24 and allowing ions to emit from the point and into the surrounding area.

In another alternative embodiment, the second dielectric layer 14 may be pre-cut with a portion removed from the top side and/or bottom side in the area where the second dielectric layer 14 will be adjacent or overtop the point 22 forming an opening 24. Therefore, when the second dielectric layer 14 is formed over the trace 16 and emitter 20, the opening 24 is adjacent the point 22, wherein the second dielectric layer 14 does not cover the point 22. In another alternative embodiment, the first dielectric layer 12 and second dielectric layer 14 both are pre-cut with the portion of the respective layer (12, 14) that may be adjacent or overtop the point 22 are removed forming openings 24. Therefore, when the trace 16 and emitter 20 are engaged to the top portion of the first dielectric layer 12, the point 22 is positioned within the opening 24, such that the first dielectric layer 12 will not cover the point 22 and the point 22 is within the opening 24. As the second dielectric layer 14 is formed over the trace 16 and emitter 20, the opening 24 is placed adjacent the point 22, so that the second dielectric layer 14 does not cover the point 22.

In the embodiment illustrated in FIGS. 2, 3, and 5, the lights 50 and conductive wire 52 are disposed on the top portion of the second dielectric layer 14. The lights 50 and conductive wire 52 may be located on any portion of the second dielectric layer 14, but as illustrated, the conductive wire 52 and the lights 50 are positioned in the space between the first trace 16 and the second trace 28. The lights 50 do not need to be obstructed by the dielectric layers (12, 14) or the traces (16, 28). Alternatively, the lights 50 and conductive wire 52 in this embodiment may be placed on the bottom portion of the first dielectric layer 12. Alternatively,

the conductive wire 52 may be positioned between the dielectric layers (12, 14) or within a dielectric layer (12, 14) with the lights 50 disposed on the top portion of the second dielectric layer 14, the bottom portion of the first dielectric layer 12 or at least visible.

An insulating jacket 27 may be positioned over at least a portion of the flexible ion generator device 10. In the cross section shown in FIG. 5, the insulating jacket 27 may encompass the first dielectric layer 12 and the second dielectric layer 14 (including trace 16, conductive wire 52 and a portion of the emitter 20 contained therein). The insulating jacket 27 may surround and protect the dielectric layers (12, 14), including the trace 16, optionally the emitter 20, and the conductive wire 52. It should be noted, the insulating jacket 27 does not cover the point 22 or obstruct the lights 50.

In another alternative embodiment and as shown in FIGS. 4, 6a, and 6b, the flexible ion generator device 10 may include a second trace 28 and a third dielectric layer 30 (in this embodiment the trace 16 becomes the first trace). The third dielectric layer 30 contains a top portion, a bottom portion, a top side, a bottom side, a left side, and a right side, wherein the top side and the bottom side are opposed to each other and the left side and the right side are opposed to each other. The conductive wire 52 may be disposed on the top portion of the third dielectric layer 30, the bottom portion of the first dielectric layer 12, or between the first dielectric layer 12 and the third dielectric layer 30 or within one of the dielectric layers (12, 14, 30).

In this embodiment, the second dielectric layer 14 and third dielectric layer 30 contain the second trace 28 disposed therebetween (and the first trace 16 is disposed between the first dielectric layer 12 and the second dielectric layer 14). The second trace 28 may be engaged on the top portion of the second dielectric layer and bottom portion of the third dielectric layer 30. The second trace 28 has a first end and a second end. The second trace 28 may be positioned on the top portion of the second dielectric layer 14. The bottom portion of the third dielectric layer 30 may be formed over the second trace 28 and coupled to the second dielectric layer 14. It will be appreciated that while the second trace 28 may be positioned on the top portion of the second dielectric layer 14, the second trace 28 may also be positioned on the bottom side of the third dielectric layer 30, and the second dielectric layer 14 is formed over the second trace 28 and coupled to the third dielectric layer 30. Furthermore, the lights 50 are engaged to the top portion of the third dielectric layer 30 or the bottom portion of the first dielectric layer 12. Preferably, the lights 50 are not obstructed so that the lights are not obstructed.

As illustrated in FIG. 4, the second trace 28 extends longitudinally along the length of the flexible ion generator device 10. In other words, the second trace 28 extends from the left side to the right side of the second dielectric layer 14 and the third dielectric layer 30. The second trace 28 contains a first end and a second end, whereby the first end is disposed adjacent the right side of the second dielectric layer 14 and third dielectric layer 30 and the second end is disposed adjacent the left side of the second dielectric layer 14 and third dielectric layer 30. The first end and the second end of the second trace 28 may extend beyond the right side and left side of the second dielectric layer 14 and third dielectric layer 30. A conductive pad or connector 18 may be disposed on the first end and/or the second end of the second trace 28. The connector 18 is engaged to a power supply for supplying power to the flexible ion electrode 10, and more specifically, the first trace 16 and second trace 28.

An emitter 20 may be engaged or etched into the second trace 28. As illustrated, the emitter 20 extends to at least the first side and/or second side of the flexible ion electrode 10. The emitter 20 extends to the top side and/or the bottom side of the flexible ion generator device 10. As shown in FIGS. 3, 4, and 7, the emitter 20 extends from the second trace 28 at an angle of about 90° to either the top side or bottom side of the flexible ion generator device 10 and away from the second trace 28. The first end of the emitter 20 is etched into the second trace 28 and the second end of the emitter 20 extends to a point 22. The point 22 of the emitter 20 may be connected, coupled, or engaged to the second dielectric layer 14 and/or the third dielectric layer 30 and extends outwards from these layers (14, 30). The point 22 may extend beyond the top side or bottom side of the dielectric layers as shown in FIGS. 1a and 1b. Alternatively, the point 22 does not extend beyond the top side or bottom side of the dielectric layers as shown in FIGS. 3 and 4.

The point 22 disposed on the second end of the emitter 20 and allows ions to flow therefrom. The second trace 28 contains at least one emitter 20, preferably at least two emitters 20, and more preferably a plurality of emitters 20. It is important to note that the second dielectric layer 14 and third dielectric layer 30 do not cover the point 22, as shown in FIG. 4, thus allowing ions to flow from the point 22 and into the surrounding area. The dielectric layers (12, 14, 30) adjacent the point may be cut-away, pulled back, or otherwise removed, allowing ions to freely flow from the end of the point 22. The emitter 20 etched into the second trace 28, extends outward from the second trace 28 and in an opposite direction than the emitter 20 that the first trace 16 extends, as shown in FIGS. 3 and 4. In other words, the emitter 20 of the first trace 16 and the emitter 20 of second trace 28 extend in opposite directions and towards opposite sides of the respective dielectric layer (12, 14, 30).

As shown in FIGS. 4, 6a, and 6b, the lights 50 and conductive wire 52 are disposed on the top portion of the third dielectric layer 30. The lights 50 and conductive wire 52 may be located on any portion of the third dielectric layer 30, but as illustrated, the conductive wire 52 and the lights 50 are positioned in the space between the first trace 16 and second trace 28. Alternatively, the lights 50 and conductive wire 52 may be placed on the bottom portion of the first dielectric layer 12. A connector 54 may be disposed on the first end and/or the second end of the conductive wire 52 and either the first end or the second end of the dielectric layers (12, 14).

As shown in FIG. 7, the emitter 20 for any of the embodiments shown in FIGS. 1-6 a,b and described herein, contains a cap 40 and an ion brush 23, containing a plurality of bristles, extending therefrom. The cap 40 is preferably composed of metal and surrounds and retains the plurality of the bristles of the ion brush 23. The cap 40 is engaged to the trace (16, 28), and electricity flows through the trace (16, 28) and into the cap 40. The electricity then flows through the cap 40 and into the plurality of the bristles of the ion brush 23. The cap 40 and the plurality of bristles of the ion brush 23 may be made of any material that conducts electricity. The cap 40 may be soldered to the trace (16, 28), allowing electrical current to flow from the trace (16, 28) and through the cap 40, and outward through the plurality of the bristles of the ion brush 23, dispensing ions from the plurality of bristles of the ion brush 23 to the surrounding area.

The bristles of the ion brush 23 are composed of a thermoplastic polymer embedded with conductive material that allows the polymer to conduct electricity. For example, the bristles of the ion brush 42 may be composed of

polypropylene or polyethylene and impregnated with carbon. Generally, the bristles of the ion brush **42** may contain between about 20 to about 80 wt % polypropylene copolymer or polyethylene copolymer, between about 5 to about 40 wt % talc, and from about 5 to 40 wt % carbon black. However, any other resistive, inductive, reactive or conductive plastic or non-metallic material may be utilized for the bristles of the ion brush **42**. The flexible ion generator device **10** may include a stiffening element within the device **10** or located at an end of one of the dielectric layers (**12**, **14**, **30**) in the embodiments shown in FIGS. 1-6. The stiffening element may include an additional dielectric layer or another device that provides additional stability or stiffens the dielectric layers (**12**, **14**, **30**).

In another alternative embodiment, only one dielectric layer may be cut-way, pulled back, or otherwise removed from the point **22** of the emitter **20**. For example and shown in FIG. 3, the second trace **28** and emitter **20** are disposed on the top portion of the second dielectric layer **14**, and the third dielectric layer **30** is formed over the second trace **28** and emitter **20** and coupled to the second dielectric layer **14**. The point **22** of the emitter **20** does not extend outward from the second dielectric layer **14** and the third dielectric layer **30**. Instead, the portion of the third dielectric layer **30** that would cover or be over top the point **22** is cut-away, pulled back, or otherwise removed, forming an opening **24** and allowing ions to emit from the point and into the surrounding area.

In another alternative embodiment, the third dielectric layer **30** may be pre-cut with a portion removed from the top side and/or bottom side in the area where the third dielectric layer **30** will be adjacent or overtop the point **22** forming an opening **24**. Therefore, when the third dielectric layer **30** is formed over the trace **16** and emitter **20**, the opening **24** is adjacent the point **22**, wherein the third dielectric layer **30** does not cover the point **22**. The second dielectric layer **14** and the third dielectric layer **30** both are pre-cut with the portion of the respective layer (**14**, **30**) that may be adjacent or overtop the point **22** are removed forming openings **24**. Therefore, when the trace **16** and emitter **20** are engaged to the top portion of the second dielectric layer **14**, the point **22** is positioned within the opening **24**, such that the second dielectric layer **14** will not cover the point **22** and the point **22** is within the opening **24**. As the third dielectric layer **30** is formed over the second trace **28** and emitter **20**, the opening **24** is placed adjacent the point **22**, so that the third dielectric layer **30** does not cover the point **22**. Additionally, both sides of the third dielectric layer **30** may have openings on the top side and bottom side, so that the third dielectric layer **30** does not cover the point **22** of the first trace **16**.

An insulating jacket **27**, as illustrated in FIG. 6b, may be positioned over at least a portion of the flexible ion generator device **10**. The insulating jacket **27** may encompass the first dielectric layer **12**, the second dielectric layer **14**, and the third dielectric layer **30** (including the first trace **16**, the second trace **28**, and a portion of the emitters **20** contained therein). The insulating jacket **27** may surround and protect the dielectric layers (**12**, **14**, **30**), including the trace **16** and emitter **20**. It should be noted, the insulating jacket **27** may surround and protect the flexible ion generator device **10**, while leaving the points **22** unobstructed for allowing the ions to flow freely and the connectors **18** to facilitate coupling of the flexible ion generator device **10** to a power supply.

In other alternative embodiments of the present invention, any number of dielectric layers may be used with or without a conducting trace in-between each dielectric layer.

Preferably, the lights **50** are ultra-violet (UV) light-emitting diode (LED) lights or UV lights. The purpose of UV lights and UV LED lights is to add the additional ability to sterilize the air, but also sterilize adjacent surfaces, such as ductwork, air handler housing, coils, filters, and the like that the flexible ion generator device **10** is adjacent.

The emitters **20** may produce negative ions or positive ions for emission into the surrounding air. For example, the embodiment illustrated in FIGS. 1, the emitters **20** may emit positive ions, negative ions, or both positive and negative ions. The emitters **20** engaged to the first trace **16** may emit positive ions and the emitters **20** engaged to the second trace **16** may emit negative ions, as shown in FIGS. 3-7.

The first dielectric layer **12** may be coated with a layer **60** composed of titanium dioxide, silver, copper or a combination thereof to create a photocatalytic reaction.

The device **10** may be positioned and secured in place within a conduit or the housing of the air handler unit, such as a duct, such that the emitters **20** are aligned generally perpendicularly to the direction of the airflow across the device, to prevent recombination of the positively charged ions with the negatively charged ions, if the flexible ion generator device **10** produces both negative and positive ions, as opposed to unipolar ionization of negative ions or positive ions.

The treatment of air by delivery of unipolar or bipolar ionization to an airflow within a conduit according to the systems and methods of the present invention may be utilized for various purposes. For example, application of bipolar ionization to an airflow within an HVAC conduit such as an air handler housing or duct may be utilized to abate allergens, pathogens, odors, gases, volatile organic compounds, bacteria, virus, mold, dander, fungus, dust mites, animal and smoke odors, and/or static electricity in a treated air space to which the airflow is directed. Ionization of air in living and working spaces may reduce building related illness and improve indoor air quality; and additionally can reduce the quantity of outside air needed to be mixed with the treated indoor air, reducing heating and cooling costs by enabling a greater degree of air recirculation.

The flexible ion generator device **10** may be used in a variable refrigerant volume (VRV) system having a shared outdoor heat exchanger and a plurality of individual air handler units. Alternatively, the HVAC system can take the form of a variable air volume (VAV), constant air volume (CAV), variable refrigerant flow (VRF) or other forms of heating, ventilation and air conditioning system.

In typical fashion, the shared outdoor heat exchanger comprises a condenser coil, compressor and fan; the individual air handler units each comprise a fan, expansion valve, heating/cooling coil(s), and a filter; and refrigerant lines connect the shared outdoor heat exchanger to the individual air handler units. Return air from the conditioned space and/or fresh air from an exterior space is treated and delivered to a conditioned air space via the individual air handler units. The outdoor heat exchanger discharges waste heat from the conditioned air space to the ambient surroundings, and/or transfers heat from a cooled zone to a heated zone. The flexible ion generator device **10** may be mounted after the filter and before the heating or cooling coil. Alternatively, the flexible ion generator device **10** may be mounted adjacent the heat exchanger located within the conduit.

Inlet airflow flows through a conduit such as the housing of the air handler unit or a duct is filtered through a filter such as a mesh, screen, paper, cloth or other filter media. A

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filtered airflow downstream of the filter is treated by discharge of bipolar ionization from the flexible ion generator device **10** to form an ionized airflow. The flexible ion generator device **10** comprises a stream of negatively charged (−) ions and/or a stream of positively charged (+). The ionized airflow enters the inlet of a fan or blower for delivery to the treated air space, and is optionally heated or cooled by passing across or through a cooling coil or heating element. The coil, filter, ion generator, and fan are optionally mounted within a housing of the air handler unit. The lights **50** shine onto the adjacent surfaces and sanitize these surfaces. Likewise, the lights **50** may sanitize the air molecules.

The length and thickness of the flexible ion generator device **10** may vary according to a number of physical or electrical parameters and desires by the user. The flexible ion generator device **10** of the present invention is provided to the user with a dielectric layer **12** having at least a first trace **14** and optionally a second trace **16**, with each trace (**14**, **16**) having an emitter **20** engaged to the trace (**14**, **16**). The user is able to cut the dielectric layer **12** to any length they desire without disrupting the performance of the device **10**. For example, if the device **10** has dielectric layers that is 5 feet in length, but the user needs a dielectric layer that is only 4 feet, the user can cut a foot off the dielectric layer without disrupting the performance of the device **10**.

An alternative embodiment of the flexible ion generator device **210** is shown in FIGS. **8-14**. The flexible ion generator device **210** includes a dielectric layer **212** with a first end, a second end, a first side, a second side, a top side, and a bottom side. A first trace **214**, and an optional second trace **216**, may be engaged to the top side of the dielectric layer **212** and extend along the top side of the dielectric layer **212** from the first end towards the second end. Periodically spaced along the length of the first trace **214** and the second trace **216** may be a contact point **218**. The contact point **218** includes a first face and a second face, wherein the first face is engaged to the top side of the dielectric layer **212**. An emitter **220** is engaged to the second face of the contact point **218**. Alternatively, the emitter **220** may be disposed on the first trace **214** and/or the second trace **216** or engaged to the first trace **214** or the second trace **216**. In another alternative embodiment, the emitters **220** on the first trace **214** may be engaged to a contact point **218** and the emitters **220** on the second trace **216** may be engaged to the second trace **216**. Similarly, in another alternative embodiment, the emitters **220** on the second trace **216** may be engaged to a contact point **218** and the emitters **220** on the first trace **214** may be engaged to the first trace **214**.

The dielectric layer **212** may be a polyamide tape, a silicon tape, or the like that has dielectric properties. One suitable tape sold by DuPont is Kapton®. The dielectric layer **212** is preferably between about 10 mils and 30 mils, and more preferably between about 15 mils to about 25 mils. The dielectric layer **212** is flexible, defined as having flexible characteristics similar to fabric, vinyl, leather, so that the dielectric layer **212** may be bent, rolled-up, twisted, contorted, deformed, misshapen, etc. The first trace **214**, and optional second trace **216**, may be formed from any substance that can conduct electricity, such as metal, nickel, gold, copper, or copper with nickel/gold plating.

The first trace **214** and the second trace **216** extend along the length of the dielectric layer **212** and substantially parallel to the first side and the second side of the dielectric layer **212**. While the first trace **214** is substantially parallel to the first side, periodically, and at a predetermined distance, the first trace **214** extends downwardly to a contact

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point **218** that is offset from the generally parallel arrangement. In other words, the first trace **214** extends along the length of the dielectric layer **212** for a predetermined distance, and at a predetermined location, the first trace **214** extends slightly towards the second side of the dielectric layer **212**. It is important that as the first trace **214** extends towards the second side of the dielectric layer **212** the first trace **214** contains a radius of curvature, meaning there are no points or sharp edges to the first trace **214**, but the first trace **214** is rounded as it extends towards the second side of the dielectric layer **212**. A contact point **218** may be positioned on the dielectric layer **212** that is offset from the parallel line the majority of the first trace **214** follows. The contact point **218** is generally square and the first trace **214** engages a first corner or first side of the contact point **218**. The first trace **214** extends from the opposite side of the contact point **218**, and preferably a second corner that is on the opposite side of the contact point **218**. The first trace **214** extends towards the first side until it reaches the parallel path and then extends parallel to the first side. Again, the first trace **214** contains a radius of curvature, meaning there are no points or sharp edges to the first trace **214**, but the first trace **214** is rounded as it extends towards the first side of the dielectric layer **212**.

In other words, the first trace **214** is positioned on the dielectric layer **212** and extends along a substantially parallel plane with respect to the first side or the second side of the dielectric layer **212** and, in predetermined locations periodically along the length of the dielectric layer **212**, the first trace **214** extends downwardly from the parallel plane for a distance and then upwardly towards the parallel plane. Likewise, the second trace **216** is positioned on the dielectric layer **212** and extends along a substantially parallel plane with respect to the first side or the second side of the dielectric layer **212** and, in predetermined locations periodically along the length of the dielectric layer **212**, the second trace **216** extends downwardly from the parallel plane for a distance and then upwardly towards the parallel plane. A contact point **218** may be placed along the first trace **214** and the second trace **216** for receiving an emitter **220** thereon. Alternatively, the emitters **220** are engaged directly to the first trace **214** and/or the second trace **216**.

The contact point **218** may be separate and apart the first trace **214**. Alternatively, the contact point **218** is integral with the first trace **214** and the first trace **214** forms the contact point **218** and is shaped as a square, rectangle, or other shape for allowing an emitter **220** to be engaged thereto. As illustrated in FIGS. **8-14**, a plurality of contact points **218** are disposed along the length of the dielectric layer **212** with each being offset, as described above, from the generally parallel path or plane, with respect to the first side, of the first trace **214**.

While the second trace **216** is substantially parallel to the second side, periodically, and at a predetermined distance, the second trace **216** extends downwardly to a contact point **218** that is offset from the generally parallel arrangement. In other words, the second trace **216** extends along the length of the dielectric layer **212** for a predetermined distance, and at a predetermined location, the second trace **216** extends slightly towards the first side of the dielectric layer **212**. It is important that as the second trace **216** extends towards the first side of the dielectric layer **212** the second trace **216** contains a radius of curvature, meaning there are no points or sharp edges to the second trace **216**, but the second trace **216** is rounded as it extends towards the first side of the dielectric layer **212**. A contact point **218** may be positioned on the dielectric layer **212** that is offset from the parallel line

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the majority of the second trace 216 follows. The contact point 218 is generally square and the second trace 216 engages a first corner or first side of the contact point 218. The second trace 216 extends from the opposite side of the contact point 218, and preferably a second corner that is on the opposite side of the contact point 218. The second trace 216 extends towards the second side until it reaches the parallel path and then extends parallel to the second side. Again, the second trace 216 contains a radius of curvature, meaning there are no points or sharp edges to the second trace 264, but the second trace 216 is rounded as it extends towards the second side of the dielectric layer 212.

As shown in FIGS. 8, 9, and 11, at least one light 250 may be disposed on the dielectric layer 212, and preferably two or more lights 250 are disposed on the dielectric layer 212, and most preferably, a plurality of lights 250 are disposed on the dielectric layer 212. As illustrated, the lights 250 are disposed on the top portion of the dielectric layer 212 but may also be disposed on the bottom portion of the dielectric layer 212. A conductive wire 252 is disposed along the dielectric layer 212 and each light 250 contacts the conductive wire 252. As illustrated, the conductive wire 252 is disposed on the top portion of the dielectric layer 212. Alternatively, the conductive wire 252 may be positioned on the bottom portion of the dielectric layer 212 or within the dielectric layer 212. The conductive wire 252 extends along the length of the dielectric layer 212 and each light 250 is engaged to the conductive wire 252. Preferably, the plurality of lights 250 are spaced an equal distance apart along the length of the conductive wire 252. The conductive wire 252 provides power to the lights 250. The conductive wire 252 is composed of metal that can conduct electricity, such as copper or aluminum.

A connector 222, similar to the connector 18 shown in FIGS. 1a and 1b, may be disposed on the first end of the first trace 214 and the second trace 216 and disposed on the first end of the dielectric layer 212. As shown in FIG. 11, a coupler 224 is disposed on the first end of the dielectric layer 212. The coupler 224 has a base portion that extends outward to an outer edge, and a first pair and a second pair of opposed sidewalls extend upwardly to an upper edge, forming a cavity therein. The intersection points of the first pair and second pair of opposed sidewalls are defined as corners. A top portion is positioned on the upper edge of the first pair and second pair of opposed sidewalls enclosing the cavity. The first pair of opposed sidewalls are longer than the second pair of opposed sidewalls, as shown in FIG. 11. The front sidewall of the first pair of opposed sidewalls facing the dielectric layer 212 contains a slit 213 therein that extends from the external side to the internal side and into the cavity of the coupler 224. The first end of the dielectric layer 212 is inserted through the slit 213 and retained within the cavity of the coupler 224.

A connector 254, similar to the connector 54 shown in FIGS. 1a and 1b, may be engaged to an end or both ends of the conductive wire 252 for engagement to a power supply for providing power to the conductive wire 252 and ultimately to the lights 250. Alternatively, the conductive wire 252 may be engaged to the connector 218 or the trace 216, and either the connector 218 or trace 216 provide the power that is ultimately transferred to the lights 250, enabling the lights 250 to illuminate. The conductive wire 252 may extend from the first end of the dielectric layer 212 to the second end of the dielectric layer 212.

As illustrated in FIG. 11, the first end of the dielectric layer 212 contains a portion of the first trace 214 and portion of the second trace 216 along with the connector 222

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engaged to the traces (214, 216) and connector 254 engaged to conductive wire 252. The connectors 222 and 254 are disposed within the cavity of the coupler 224. The back sidewall of the first pair of opposed sidewalls facing away from the dielectric layer 212 contains three bores for receiving a first wire 226, a second wire 228, and a third wire 256 from the power supply device 230. The first wire 226 contacts the connector 222 engaged to the first trace 214 for supplying power to the first trace 214. The second wire 228 is engaged to the connector 222 engaged to the second trace 216 for supplying power to the second trace 216. The third wire 256 is engaged to the connector 254 engaged to the conductive wire 252 for supplying power to the conductive wire 252.

In the embodiments illustrated in FIGS. 12-14, there is only one wire 226 extending from the power supply device 230 and into the coupler 224. In this embodiment, the coupler 224 only has one bore for receiving the one wire 226. The wire 226 is engaged to the connector 222 of the first trace 214 in these embodiments for supplying power to the first trace 214. The wire 226 is also engaged to the connector 254 of the conductive wire 252 for supplying power to the lights 250.

The power supply device 230 provides alternating current or direct current, with constant or varying frequency, constant or varying voltage and constant or varying current. As illustrated in FIG. 11, the power supply device 230 houses a high voltage transformer 232, a microprocessor 234, and a light emitting-diode (LED) 236. As illustrated, the power supply device 230 is a box, having a base portion that extends outwards to an outer edge and a first pair and a second pair of opposed sidewalls extending upward therefrom to an upper edge, forming a cavity therein. A top portion is engaged to the upper edge and enclosing the cavity. The high voltage transformer 232 and the microprocessor 234 are disposed within the cavity of the power supply device 230. The LED 236 is visible when the top portion is engaged. As illustrated in FIGS. 8 and 9, the top portion contains an opening 238 wherein the LED 236 may extend through the opening 238 or at least be visible through the opening 238. The LED 236 is communicatively coupled to the microprocessor 234, wherein when the device 210 is operating effectively as indicated by the microprocessor, a signal is sent by the microprocessor 234 that illuminates the LED 236. If the device 210 is not operating properly, the microprocessor 234 does not send the signal and the LED 236 is not illuminated. Alternatively, the microprocessor 234 sends a signal to illuminate the LED 236 when the device 210 is not operating properly and no signal is sent when the device 210 is operating properly. Alternatively, the LED 236 may be illuminated by a signal sent by the microprocessor 234 when power is flowing from the power supply device 230 and into the traces (214, 216) and emitting ions.

As shown in FIG. 8, the emitter 220 contains a cap 240 and an ion brush 242, containing a plurality of bristles, extending therefrom. The cap 240 is preferably composed of metal and surrounds and retains the plurality of the bristles of the ion brush 242. The cap 240 is engaged to the contact point 218, and electricity flows through the contact point 218 and into the cap 240. The electricity then flows through the cap 240 and into the plurality of the bristles of the ion brush 242. The cap 240 and the plurality of bristles of the ion brush 242 may be made of any material that conducts electricity. The cap 240 may be soldered to the contact point 218, allowing electrical current to flow from the trace (214, 216), into the contact point 218, and then through the cap 240, and outward through the plurality of the bristles of the ion brush

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242, dispensing ions from the plurality of bristles of the ion brush **242** to the surrounding area.

In one embodiment, the bristles of the ion brush **242** are composed of a thermoplastic polymer imbedded with conductive material that allows the polymer to conduct electricity. For example, the bristles of the ion brush **242** may be composed of polypropylene or polyethylene and impregnated with carbon. Generally, the bristles of the ion brush **242** may contain between about 20 to about 80 wt % polypropylene copolymer or polyethylene copolymer, between about 5 to about 40 wt % talc, and from about 5 to 40 wt % carbon black. However, any other resistive, inductive, reactive or conductive plastic or non-metallic material may be utilized for the bristles of the ion brush **242**. The flexible ion generator device **210** may include a stiffening element within the device **210** or located at an end of one of the dielectric layer **212**. The stiffening element may include an additional dielectric layer or another device that provides additional stability or stiffens the dielectric layer **212**.

The dielectric layer **212** may be coated with a layer **260** composed of titanium dioxide, silver, copper or a combination thereof to create a photocatalytic reaction.

In another alternative embodiment, the emitter **220** may have a first end and a second end, wherein the first end of the emitter **220** is etched into the first trace **214** and/or the second trace **216** and extends to a point. The second end of the emitter **220** has a gradually reducing width that terminates at a sharp point, allowing ions to flow therefrom. The point of the emitter **220** is not connected, coupled, or engaged to the dielectric layer **212** and may extend outwards from the dielectric layer **212**. In other words, the emitter **220** and/or point of the emitter **220** may extend beyond the top side or bottom side of the dielectric layer **212**. The point may be coated or plated with a corrosion resistant layer such as gold or other coating material.

The point disposed on the second end of the emitter **220** allows ions to flow therefrom. It is important to note that the dielectric layer **212** does not cover the point, thus allowing ions to flow from the point and into the surrounding air. The dielectric layer **212** adjacent the point may be cut-away, pulled back, or otherwise removed, allowing ions to freely flow from the end of the point.

The emitters **220** may produce negative ions or positive ions for emission into the surrounding air. For example, the embodiment illustrated in FIG. 8, the emitters **220** engaged to the trace **214** or the contact points **218** of the first trace **214** may emit positive ions and the emitters **220** engaged to the second trace **216** or the contact points **218** of the second trace **216** may emit negative ions. In the embodiment illustrated in FIG. 12, the emitters **220** emit either positive or negative ions, or a combination thereof.

The device **210** may be positioned and secured in place within a conduit or the housing of the air handler unit, such as a duct, such that the emitters **220** are aligned generally perpendicularly to the direction of the airflow across the device, to prevent recombination of the positively charged ions with the negatively charged ions, if the flexible ion electrode **210** produces both negative and positive ions, as opposed to unipolar ionization of negative ions or positive ions.

The device **210** may be positioned in place or engaged to an air handler unit or like device by a flexible ion generator attachment device **244** and the power supply device **230** may contain a power supply device attachment device **246**. As shown in FIGS. 9, 10-12, the flexible ion generator attachment device **244** is engaged to the bottom side of the dielectric layer **212**. Likewise, the power supply device

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attachment device **246** is engaged to the external side of the base of the power supply device **230**. The flexible ion generator attachment device **244** and the power supply device attachment device **246** may be double sided tape, wherein one side of the tape is engaged to the bottom side of the dielectric layer **212** and the external side of the base of the power supply device **230**. The second side of the double sided tape is engaged to the place, air handler unit, or other location where the device **210** is desired to be engaged. Alternatively, the flexible ion generator attachment device **244** and the power supply device attachment device **246** may be a hook and loop fastener, commonly sold under the trademark Velcro®, wherein the hook portion is engaged to the bottom side of the dielectric layer **212** and the external side of the base of the power supply device **230**, and the loop portion is engaged to the place, air handler unit, or other location where the device **210** is desired to be engaged, for selectively securing the device **210**. Alternatively, the loop portion is engaged to the bottom side of the dielectric layer **212** and the external side of the base of the power supply device **230**, and the hook portion is engaged to the place, air handler unit, or other location where the device **210** is desired to be engaged, for selectively securing the device **210**.

As illustrated in FIG. 13 in an alternative embodiment, the emitters **220** disposed along the first trace **214** may face opposite each other. In other words, the emitters **220** may alternate positions along the length of the first trace **214**. As shown, the first emitter **220** faces towards one side, the second emitter **220'** faces towards the opposite side, the third emitter **220''** faces towards the same side as emitter **220**, and the fourth emitter **220'''** faces towards the same side as emitter **220'** and so on.

As illustrated in FIG. 14, the emitters **220** may be positioned where they extend upwardly from the dielectric layer **212**. The cap **240** is positioned on the contact point **218** or the trace **214** and the ion brush **242** with its plurality of bristles extends upwardly from the dielectric layer **212** and not laying on the dielectric layer **212** or in parallel with the dielectric layer **212** as shown in FIGS. 8-12.

The treatment of air by delivery of unipolar or bipolar ionization to an airflow within a conduit according to the systems and methods of the present invention may be utilized for various purposes. For example, application of bipolar ionization to an airflow within an HVAC conduit such as an air handler housing or duct may be utilized to abate allergens, pathogens, odors, gases, volatile organic compounds, bacteria, virus, mold, dander, fungus, dust mites, animal and smoke odors, and/or static electricity in a treated air space to which the airflow is directed. Ionization of air in living and working spaces may reduce building related illness and improve indoor air quality; and additionally can reduce the quantity of outside air needed to be mixed with the treated indoor air, reducing heating and cooling costs by enabling a greater degree of air recirculation.

The flexible ion generator device **210** may be used in a variable refrigerant volume (VRV) system having a shared outdoor heat exchanger and a plurality of individual air handler units. Alternatively, the HVAC system can take the form of a variable air volume (VAV), constant air volume (CAV), variable refrigerant flow (VRF) or other forms of heating, ventilation and air conditioning system.

In typical fashion, the shared outdoor heat exchanger comprises a condenser coil, compressor and fan; the individual air handler units each comprise a fan, expansion valve, heating/cooling coil(s), and a filter; and refrigerant

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lines connect the shared outdoor heat exchanger to the individual air handler units. Return air from the conditioned space and/or fresh air from an exterior space is treated and delivered to a conditioned air space via the individual air handler units. The outdoor heat exchanger discharges waste heat from the conditioned air space to the ambient surroundings, and/or transfers heat from a cooled zone to a heated zone. The flexible ion generator device **210** may be mounted after the filter and before the heating or cooling coil. Alternatively, the flexible ion generator device **210** may be mounted adjacent the heat exchanger located within the conduit.

Inlet airflow flows through a conduit such as the housing of the air handler unit or a duct is filtered through a filter such as a mesh, screen, paper, cloth or other filter media. A filtered airflow downstream of the filter is treated by discharge of bipolar ionization from the flexible ion generator device **210** to form an ionized airflow. The flexible ion generator device **210** comprises a stream of negatively charged (−) ions and/or a stream of positively charged (+) ions. The ionized airflow enters the inlet of a fan or blower for delivery to the treated air space, and is optionally heated or cooled by passing across or through a cooling coil or heating element. The coil, filter, ion generator, and fan are optionally mounted within a housing of the air handler unit.

The length and thickness of the flexible ion generator device **210** may vary according to a number of physical or electrical parameters and desires by the user. The flexible ion generator device **210** of the present invention is provided to the user with a dielectric layer **212** having at least a first trace **214** and optionally a second trace **216**, with each trace (**214**, **216**) having either a contact point **218** and emitter **220** or only an emitter **220** engaged to the trace (**214**, **216**). The user is able to cut the dielectric layer **212** to any length they desire without disrupting the performance of the device **210**. For example, if the device **210** has dielectric layers that is 5 feet in length, but the user needs a dielectric layer that is only 4 feet, the user can cut a foot off the dielectric layer without disrupting the performance of the device **210**.

Preferably, the lights **250** are ultra-violet (UV) light-emitting diode (LED) lights or UV lights. The purpose of UV lights and UV LED lights is to add the additional ability to sterilize the air, but also sterilize adjacent surfaces, such as ductwork, air handler housing, coils, filters, and the like that the flexible ion generator device **210** is adjacent.

Although the present invention has been illustrated and described herein with reference to preferred embodiments and specific examples thereof, it will be readily apparent to those of ordinary skill in the art that other embodiments and examples may perform similar functions and/or achieve like results. All such equivalent embodiments and examples are within the spirit and scope of the present invention and are intended to be covered by the following claims.

What is claimed is:

1. A flexible ion generator device, comprising:
 - a dielectric layer having a top side and a bottom side;
 - at least one trace having a first end and a second end, the at least one trace is engaged to the top side of the dielectric layer;
 - at least one emitter engaged to the trace for emitting ions; and
 - an attachment device engaged to the bottom side of the dielectric layer.
2. The flexible ion generator electrode according to claim 1, wherein the emitter has a first end and a second end, the first end of the emitter is engaged to the trace and the second end extends away from the trace.

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3. The flexible ion generator device according to claim 1, wherein the dielectric layer is composed of polyamide.

4. The flexible ion generator device according to claim 1, wherein the emitter contains a plurality of bristles for emitting ions.

5. The flexible ion generator device according to claim 1, further comprising a connector engaged to the first end of the trace.

6. The flexible ion generator device according to claim 1, further comprising a power supply engaged to the flexible ion generator device.

7. The flexible ion generator device according to claim 6, wherein the power supply contains a light-emitting diode (LED).

8. A flexible ion generator device, comprising:

- at least one dielectric layer having a top portion, a bottom portion, a top side, a bottom side, a left side, and a right side;

- at least one trace having a first end and a second end, the at least one trace is engaged to the top side of one of the dielectric layers;

- a plurality of emitters engaged to the trace for emitting ions; and

- an attachment device engaged to the bottom side of one of the dielectric layers.

9. The flexible ion generator device according to claim 8, wherein the emitter has a first end and a second end, the first end of the emitter is engaged to the at least one trace and the second end extends away from the trace.

10. The flexible ion generator device according to claim 8, wherein the dielectric layer is composed of polyamide.

11. The flexible ion generator device according to claim 8, wherein the emitter contains a plurality of bristles for emitting ions.

12. The flexible ion generator device according to claim 8, further comprising a connector engaged to the first end of the trace.

13. The flexible ion generator device according to claim 12, further comprising a power supply engaged to the connector.

14. The flexible ion generator device according to claim 8, wherein the attachment device is a hook and loop fastener.

15. The flexible ion generator device according to claim 8, wherein the device is mounted inside a VRV/VRF air handling system.

16. A method of producing ions, comprising:

- providing a flexible ion generator device comprising dielectric layer having a top side and a bottom side, at least one trace having a first end and a second end, wherein the at least one trace is engaged to the top side of the dielectric layer, at least one emitter engaged to the trace for emitting ions, and an attachment device engaged to the bottom side of the dielectric layer;

- mounting the flexible ion generator device within a conduit with the attachment device; and
- positioning the at least one emitter within airflow within the conduit.

17. The method of producing ions according to claim 16, further comprising a connector engaged to the first end of the trace and a power supply engaged to the connector.

18. The method of producing ions according to claim 16, further comprising mounting the flexible ion generator device after a prefilter and before a cooling coil.

19. The method of producing ions according to claim 16, further comprising mounting the flexible ion generator device to a conduit using the attachment device.

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20. The method of producing ions according to claim **16**, wherein the flexible ion generator device is mounted inside a VRV/VRF air handling system.

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